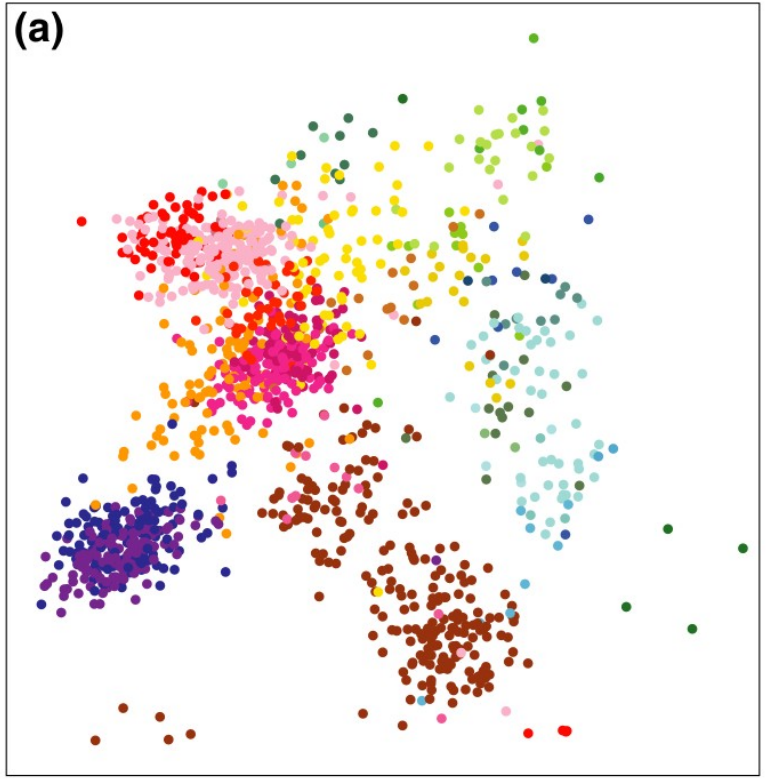
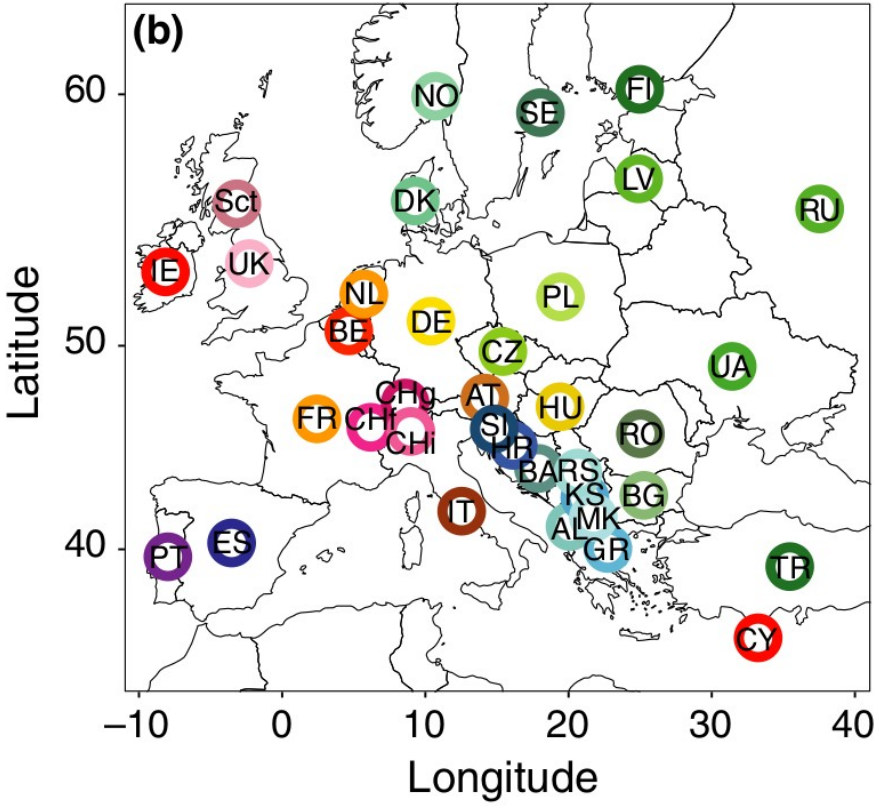
PCA for Population Structure Analysis

Nikolay Oskolkov, Group Leader (PI) at NIRI, 12.11.2026





Document outline

- Increased burden of ultra-rare structural variants localizing to boundaries of topologically associated domains in schizophrenia
  - Results
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    - Burden analysis of ultra-rare SVs
    - Validation and analysis of ultra-rare TADs-affecting SVs
    - Common variants with large effects were not identified
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    - Burden of ultra-rare SNV/Indels
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    - Single-variant association analysis

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ARTICLE

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https://doi.org/10.1038/s41467-020-15707-w OPEN

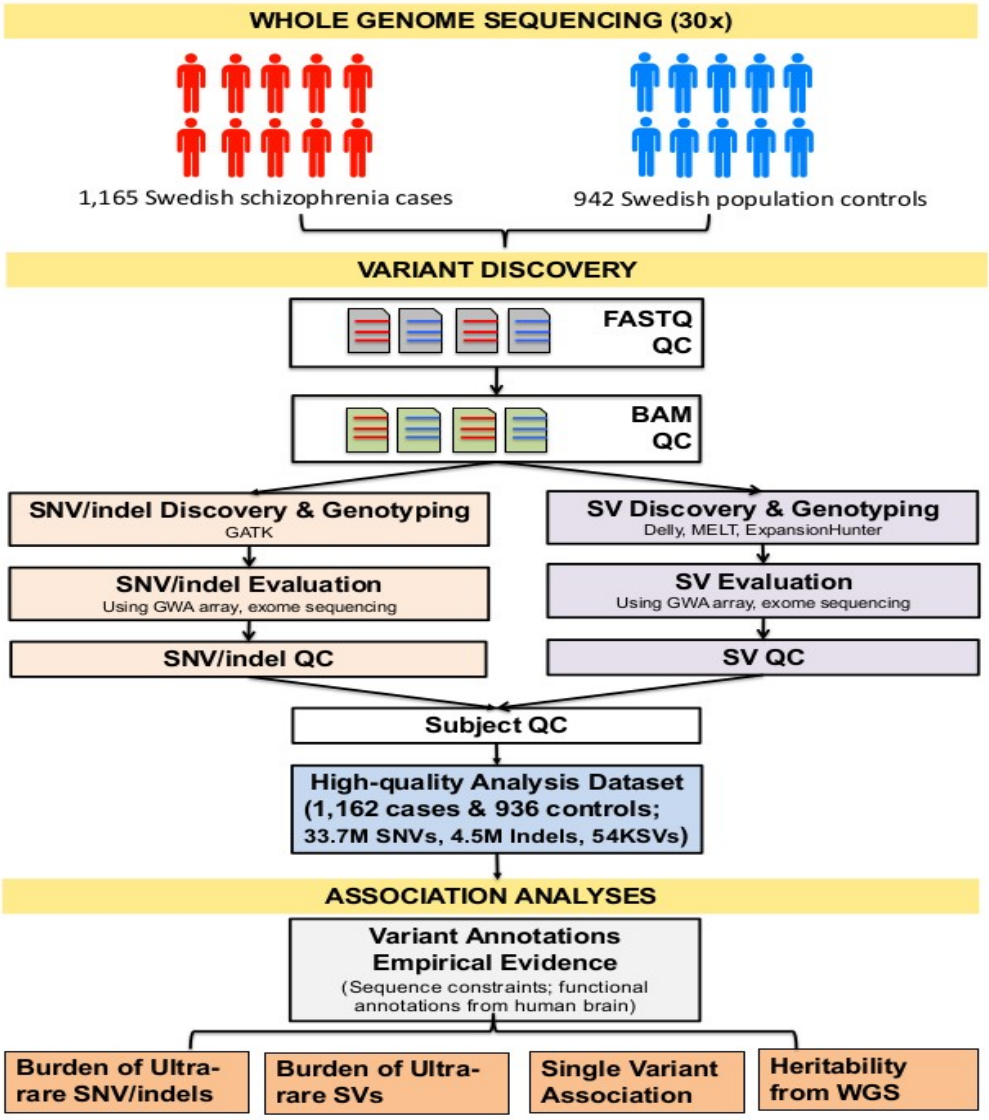
Increased burden of ultra-rare structural variants localizing to boundaries of topologically associated domains in schizophrenia

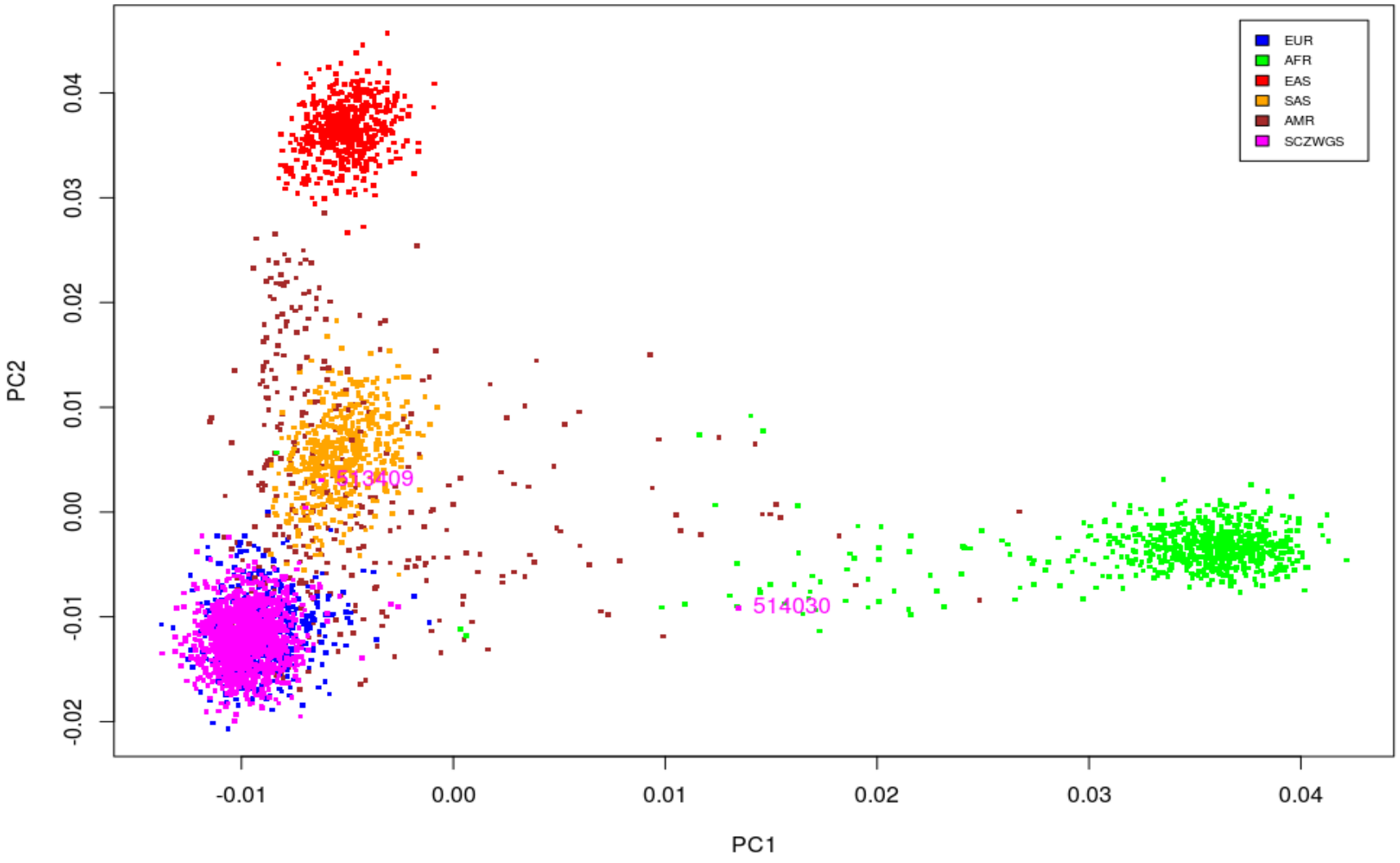
Matthew Halvorsen[1,10], Ruth Huh[2,10], Nikolay Oskolkov[3,10], Jia Wen[1,10], Sergiu Netotea[4], Paola Giusti-Rodriguez[1], Robert Karlsson[5], Julien Bryois[5], Björn Nystedt[6], Adam Ameur[7], Anna K. Kähler[5], NaEshia Ancalade[1], Martilias Farrell[1], James J. Crowley[1,8,9], Yun Li[1,2], Patrik K. E. Magnusson[5], Ulf Gyllensten[7], Christina M. Hultman[5], Patrick F. Sullivan[1,5,8✉] & Jin P. Szatkiewicz[1,8✉]

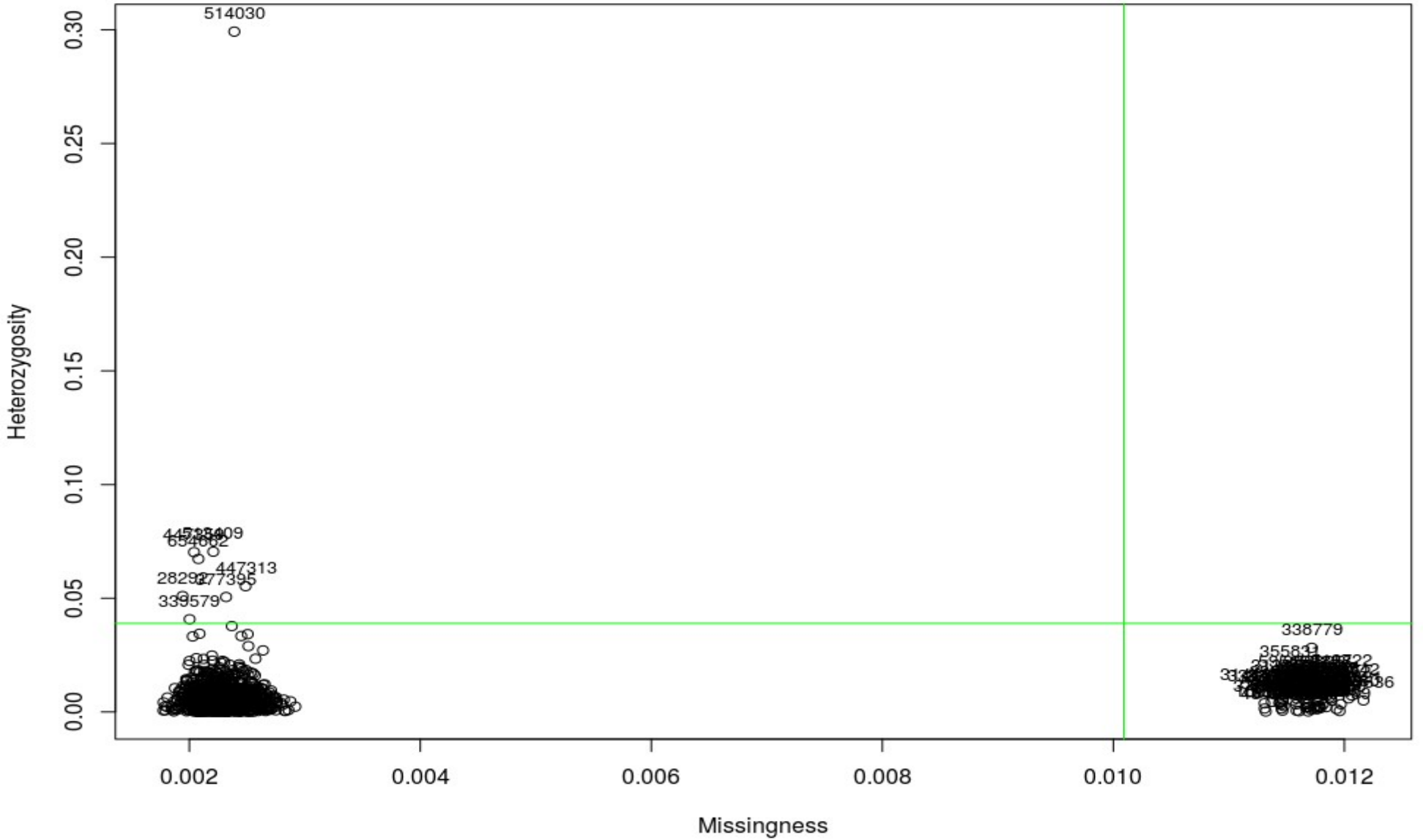
Despite considerable progress in schizophrenia genetics, most findings have been for large rare structural variants and common variants in well-imputed regions with few genes implicated from exome sequencing. Whole genome sequencing (WGS) can potentially provide a more complete enumeration of etiological genetic variation apart from the exome and regions of high linkage disequilibrium. We analyze high-coverage WGS data from 1162 Swedish schizophrenia cases and 936 ancestry-matched population controls. Our main objective is to evaluate the contribution to schizophrenia etiology from a variety of genetic variants accessible to WGS but not by previous technologies. Our results suggest that ultra-rare structural variants that affect the boundaries of topologically associated domains (TADs) increase risk for schizophrenia. Alterations in TAD boundaries may lead to dysregulation of gene expression. Future mechanistic studies will be needed to determine the precise functional effects of these variants on biology.

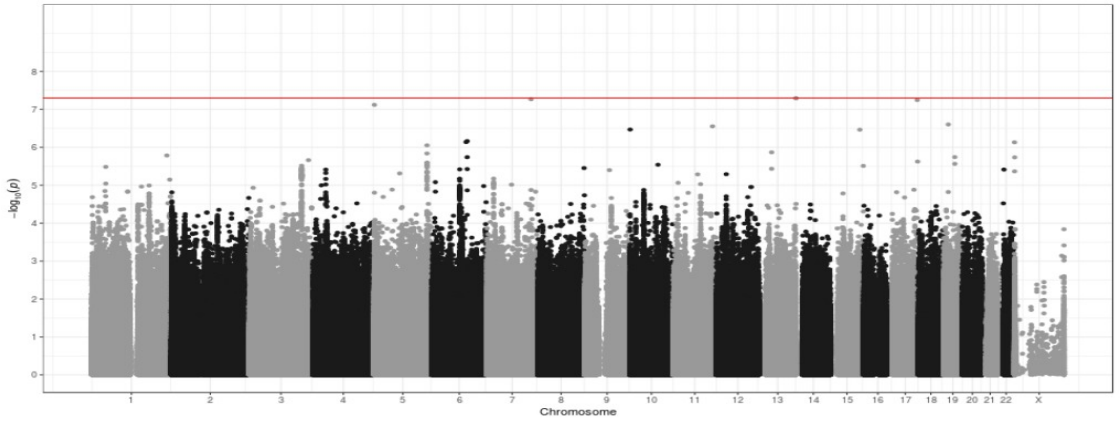
[1] Department of Genetics, University of North Carolina, Chapel Hill, NC 27599, USA. [2] Department of Biostatistics, University of North Carolina, Chapel Hill, NC 27599, USA. [3] Department of Biology, National Bioinformatics Infrastructure Sweden, Science for Life Laboratory, Lund University, 22362 Lund, Sweden. [4] Department of Biology and Biological Engineering, National Bioinformatics Infrastructure Sweden, Science for Life Laboratory, Chalmers University of Technology, 41258 Göteborg, Sweden. [5] Department of Medical Epidemiology and Biostatistics, Karolinska Institutet, 17177 Stockholm, Sweden. [6] Department

Schizophrenia is heritable with 60-80% of estimated heritability, hence GWAS, focus on contribution from rare and ultra-rare variants (potentially causal)

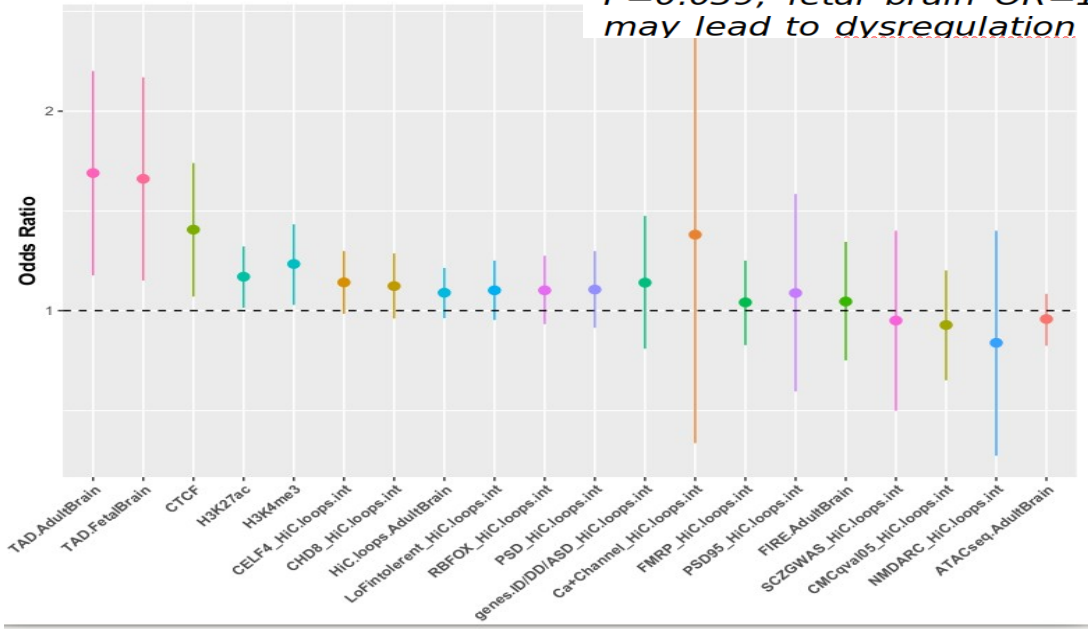


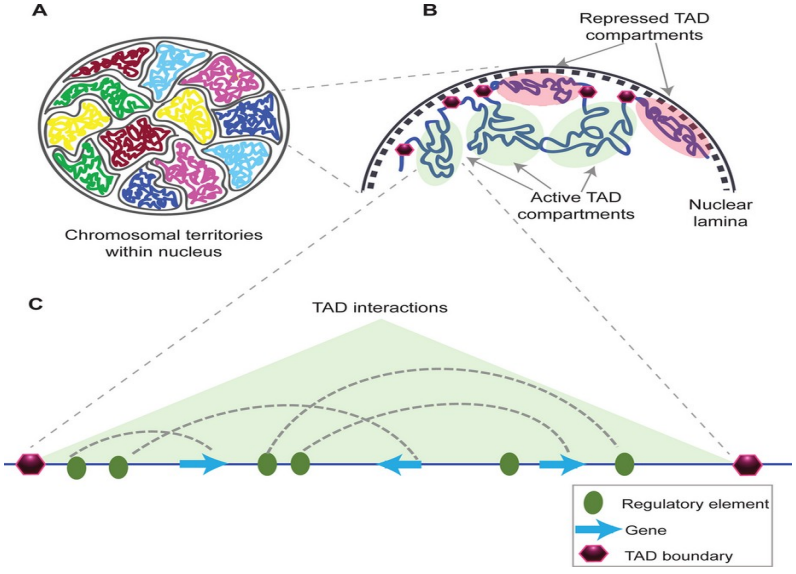

1000G+SCZWGS
EUR
AFR
EAS
SAS
AMR
SCZWGS
513409
514030
PC1
PC2
-0.01
0.00
0.01
0.02
0.03
0.04
-0.02
-0.01
0.00
0.01
0.02
0.03
0.04


514030
447313
339579
338779
Heterozygosity
Missingness
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.002
0.004
0.006
0.008
0.010
0.012



interactions (OR=1.028, adjusted P=0.028). Furthermore, our results suggest that ultra-rare structural variants that affect the boundaries of topologically associated domains (TADs) increase risk for schizophrenia (adult brain OR=1.61, adjusted P=0.039; fetal brain OR=1.58, adjusted P=0.039). Alterations in TAD boundaries may lead to dysregulation of gene expression. As in a recent report on height and





PCA Introduction

How to compute PCA from scratch

Coding in R:

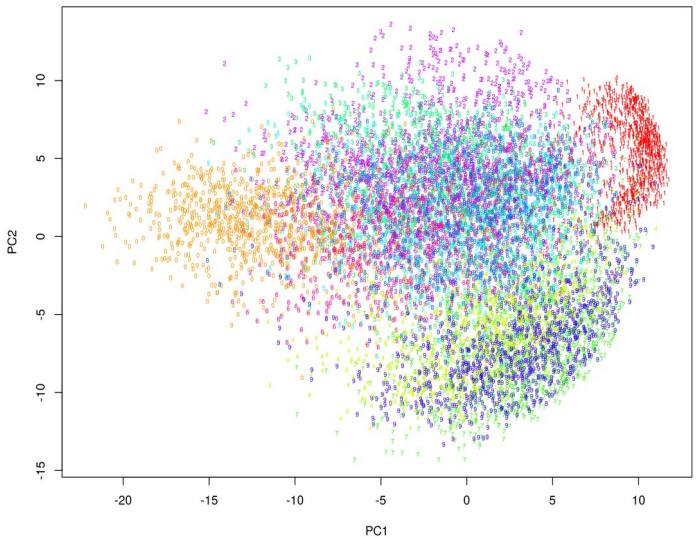
Mathematically:

```
data_centered <- scale(data, center = TRUE, scale = FALSE)
```
→ $M_{ij} = X_{ij} - \mu_j$

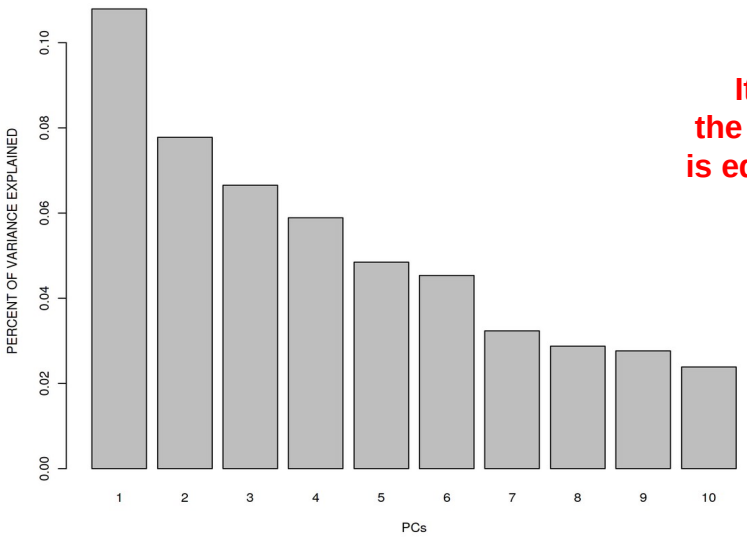
```
covariance <- t(data_centered) %*% data_centered
```
→ $A = (1/N)*M^T*M$

```
eig <- eigen(covariance)
```
→ $A*u = \lambda*u$

```
plot(eig$vectors[,1:2]);
```




```
barplot(eig$values / sum(eig$values))
```



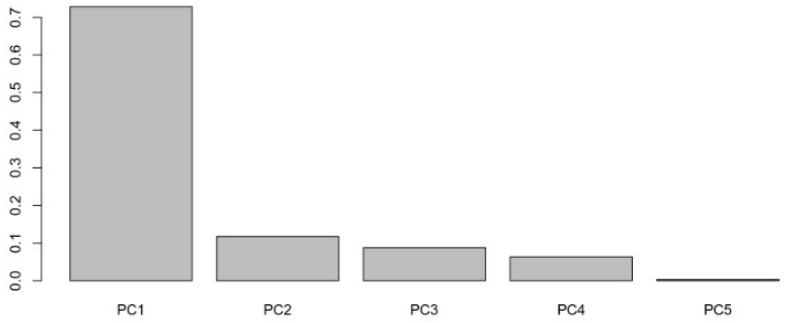
It can be analytically derived that the eigen value decomposition in PCA is equivalent to projecting data on axes of maximal variation in the data

Toy example of PCA for population genetics

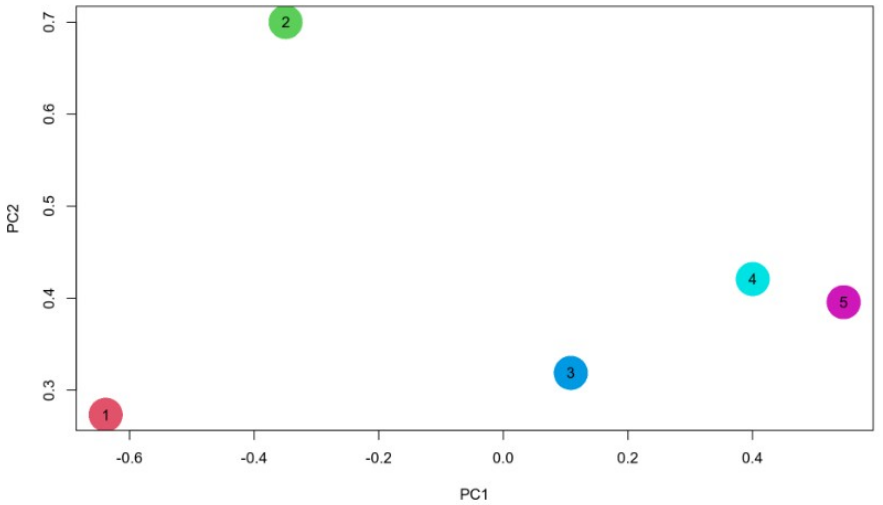
```r
gen <- t(matrix(c(1, 0, 2, 0, 2, 0, 2, 1, 1, 1, 0, 1, 0, 2, 1, 2, 1, 1, 1, 1, 1,
    0, 1, 0, 2, 0, 1, 1, 0, 2, 1, 2, 0, 1, 0), 5, by = TRUE))
colnames(gen) <- paste0("Ind", 1:5)
rownames(gen) <- paste0("SNP", 1:7)
print(gen)
```

```
     Ind1 Ind2 Ind3 Ind4 Ind5
SNP1    1    1    1    0    0
SNP2    0    1    2    1    2
SNP3    2    1    1    0    1
SNP4    0    0    1    2    2
SNP5    2    1    1    0    0
SNP6    0    0    1    1    1
SNP7    2    2    1    1    0
```

```r
gen_centered <- scale(gen, center = TRUE, scale = FALSE)
covariance <- t(gen_centered) %*% gen_centered
eig <- eigen(covariance)
barplot(eig$values/sum(eig$values), names = paste0("PC", 1:5))
```



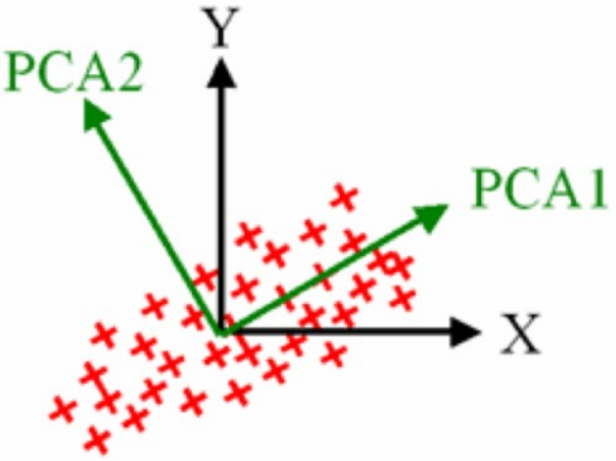

```r
plot(eig$vectors[, 1:2], xlab = "PC1", ylab = "PC2", pch = 16, cex = 5, col = 2:6)
points(eig$vectors[, 1:2], pch = as.character(1:5))
```



Idea: project the data into a low dimensional space that explains the largest amount of variance

How maximization of covariance is related to eigen value decomposition

- Collapse p features (p>>n) to few latent features and keep variation
- Rotation and shift of coordinate system toward maximal variance
- PCA is an **eigen matrix decomposition** problem



$$PC = u^T X = X^T u$$

X is mean centered $\Longrightarrow < PC >= 0$

$$< (PC - < PC >)^2 >=< PC^2 >= u^T X X^T u$$

$$XX^T = A$$

$$< PC^2 >= u^T A u$$

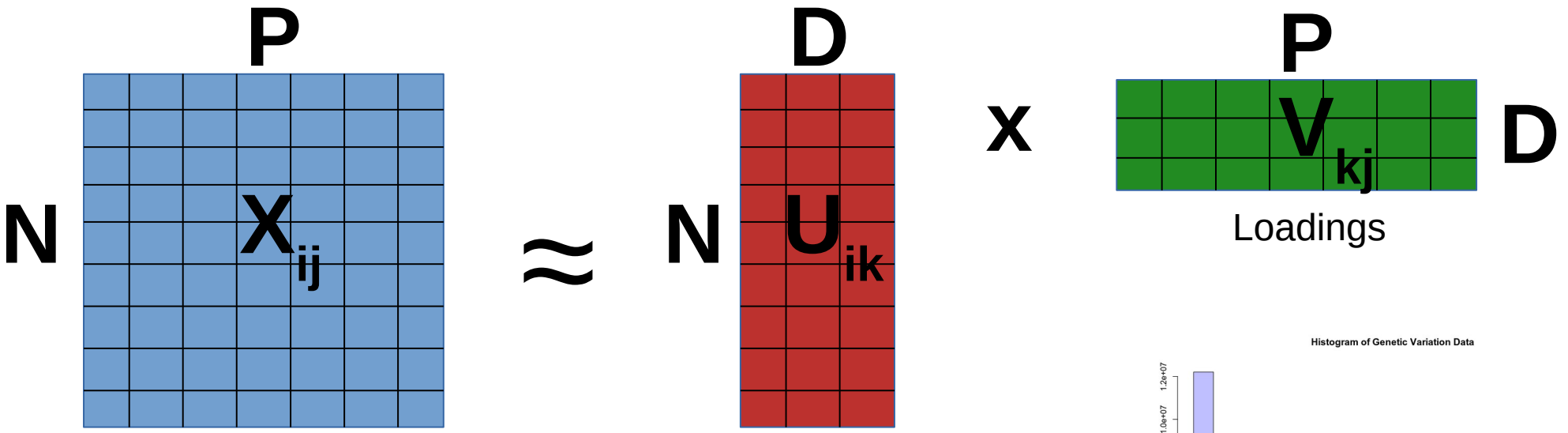
A is **variance-covariance** of X

$$\max(u^T A u + \lambda(1 - u^T u)) = 0$$

$$Au = \lambda u$$

PCA on Genetic Variation: Euclidean Distance

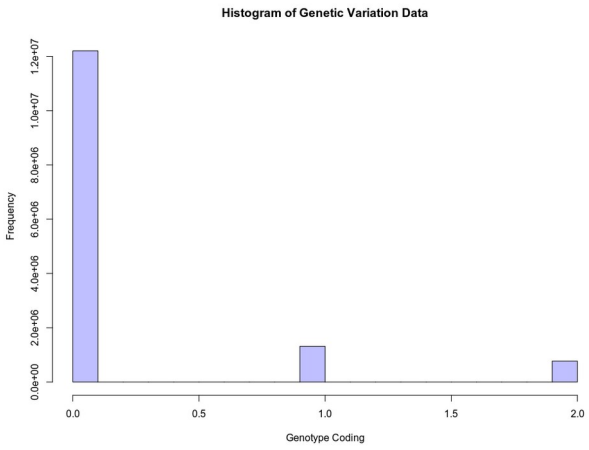
$$X_{ij} \approx U_{ik} V_{kj}$$



Data

Low-dimensional data representation (embeddings)

Loadings



$$\text{Loss} = \sum_{i=1}^{N} \sum_{j=1}^{P} (X_{ij} - U_{ik} V_{kj})^2$$

PCA and Gaussian Distribution Assumption

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Population Structure and Eigenanalysis

Nick Patterson[1*], Alkes L. Price[1,2], David Reich[1,2]

1 Broad Institute of Harvard and MIT, Cambridge, Massachusetts, United States of America, **2** Department of Genetics, Harvard Medical School, Boston, Massachusetts, United States of America

Current methods for inferring population structure from genetic data do not provide formal significance tests for population differentiation. We discuss an approach to studying population structure (principal components analysis) that was first applied to genetic data by Cavalli-Sforza and colleagues. We place the method on a solid statistical footing, using results from modern statistics to develop formal significance tests. We also uncover a general "phase change" phenomenon about the ability to detect structure in genetic data, which emerges from the statistical theory we use, and has an important implication for the ability to discover structure in genetic data: for a fixed but large dataset size, divergence between two populations (as measured, for example, by a statistic like F_{ST}) below a threshold is essentially undetectable, but a little above threshold, detection will be easy. This means that we can predict the dataset size needed to detect structure.

Citation: Patterson N, Price AL, Reich D (2006) Population structure and eigenanalysis. PLoS Genet 2(12): e190. doi:10.1371/journal.pgen.0020190

Introduction

A central challenge in analyzing any genetic dataset is to explore whether there is any evidence that the samples in the data are from a population that is structured. Are the individuals from a homogeneous population or from a population containing subgroups that are genetically distinct? Can we find evidence for substructure in the data, and quantify it?

This question of detecting and quantifying structure arises in medical genetics, for instance, in case-control studies where uncorrected population structure can induce false positives [1]. It also arises in population genetics, where understanding of the structure may be important to the key scientific issues, especially uncovering the demographic history of the population under study.

We focus on principal components analysis (PCA), which was first introduced to the study of genetic data almost thirty practical, even on the largest datasets. This is our main aim in this paper.

Using some recent results in theoretical statistics, we introduce a formal test statistic for population structure. We also discuss testing for *additional* structure after some structure has been found. Finally, we are able to estimate the degree of population differentiation that will be detectable for a given data size.

Our methods work in a broad range of contexts, and can be modified to work with markers in linkage disequilibrium (LD). The methods are also able to find structure in admixed populations such as African Americans—that is, in which individuals inherit ancestry from multiple ancestral populations—as long as the individuals being studied have different proportional contributions from the ancestral populations.

We believe that principal components methods largely fell out of favor with the introduction of the sophisticated cluster-based program STRUCTURE [9,10]. STRUCTURE

One concern with applying this approach to genetic data is that the entries in the matrix M do not have the Gaussian distributions expected for a Wishart matrix; instead, they correspond to the three possible genotypes at each SNP. However it is not critical that the entries in the m x n matrix M be Gaussian. Soshnikov [25] showed that the same TW limit arose if the cell entries were any distribution with high-order moments no greater than the Gaussian. The matrix X is a sum of n rank 1 matrices, and Soshnikov's result suggests that the same limit would be obtained from any probability distribution in which the columns of M are independent, isotropic (all directions are equiprobable), and such that the column norms have moments no larger than those for a column of independent Gaussian entries. In all our genetic applications, the column norms are in fact bounded, so we can expect the sample covariance matrices to behave well. This theory, originally developed for the case of Gaussian matrix entries, thus seemed likely to work well with large genetic biallelic data arrays. The remainder of this paper verifies that this is the case.

What Flora Jay thinks about non-Gaussian distribution of genetic variation data

Both PCA and FA optimize a Euclidean distance (L2 norm) between the original data and its low-rank approximation. L2 norm is a consequence of data normality assumption. Why do people still use PCA / FA for non-normally distributed categorical data such as genetic variation?

This norm is often used for convenience because it allows easy and quick/immediate optimization of the problem. So it is true that the loss/ or the model could be changed. There is a trade off between the "quality/accuracy/reality" of the model and the optimization procedure. What we personally experimented with, rather than changing the loss was to add information in the form of constraints on the U or V matrices (e.g. Caye et al https://hal.archives-ouvertes.fr/hal-01676712). It works but leads to more complex non deterministic optimization procedure. For other models you can think about Dystruct, Admixture, treemix etc.

Toy example of MDS for population genetics

```r
gen <- t(matrix(c(1, 0, 2, 0, 2, 0, 2, 1, 1, 1, 0, 1, 0, 2, 1, 2, 1, 1, 1, 1, 1,
    0, 1, 0, 2, 0, 1, 1, 0, 2, 1, 2, 0, 1, 0), 5, by = TRUE))
colnames(gen) <- paste0("Ind", 1:5)
rownames(gen) <- paste0("SNP", 1:7)
print(gen)
```

```
     Ind1 Ind2 Ind3 Ind4 Ind5
SNP1    1    1    1    0    0
SNP2    0    1    2    1    2
SNP3    2    1    1    0    1
SNP4    0    0    1    2    2
SNP5    2    1    1    0    0
SNP6    0    0    1    1    1
SNP7    2    2    1    1    0
```

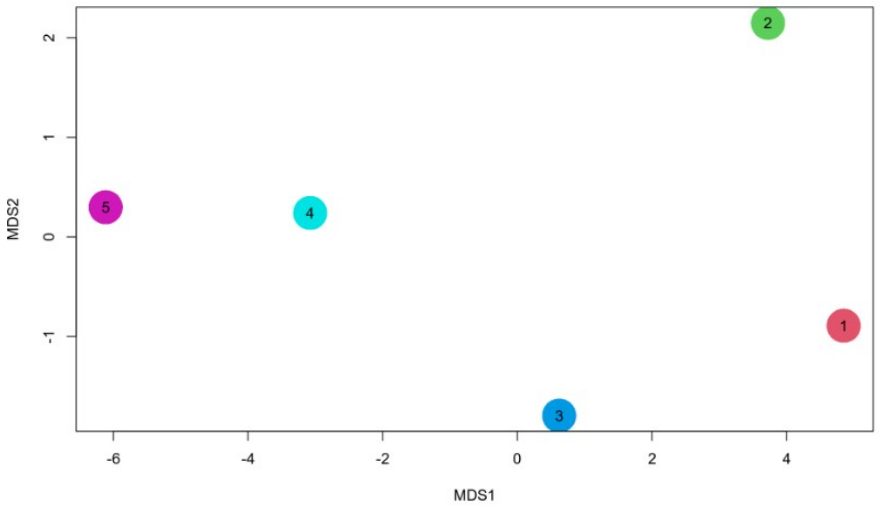
```r
my_dist <- dist(t(gen), method = "manhattan", upper = TRUE, diag = TRUE)
my_dist
```

```
     Ind1 Ind2 Ind3 Ind4 Ind5
Ind1    0    3    7   10   11
Ind2    3    0    4    7    8
Ind3    7    4    0    5    4
Ind4   10    7    5    0    3
Ind5   11    8    4    3    0
```

```r
my_mds <- cmdscale(my_dist)
my_mds
```

```
          [,1]       [,2]
Ind1 -6.114866  0.2977272
Ind2 -3.077094  0.2402932
Ind3  0.621230 -1.7944048
Ind4  3.723118  2.1483083
Ind5  4.847612 -0.8919238
```

```r
plot(my_mds, xlab = "MDS1", ylab = "MDS2", pch = 16, cex = 5, col = 6:2)
points(my_mds, pch = as.character(5:1))
```



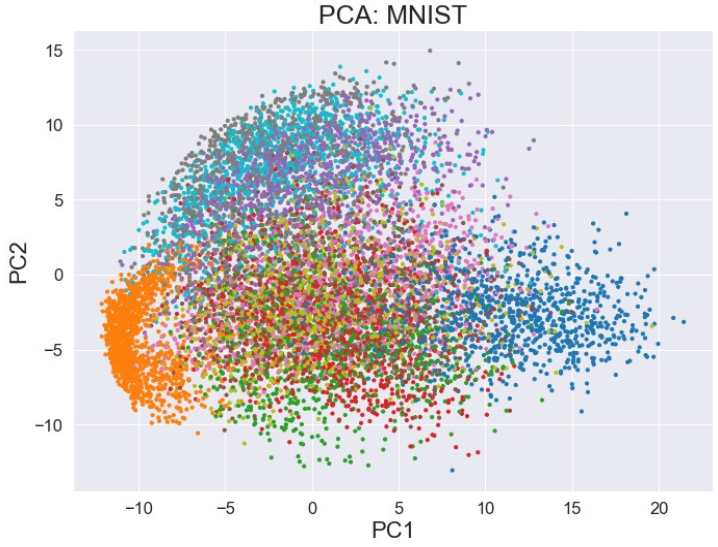
Idea: project the data into a low dimensional space that preserves distances

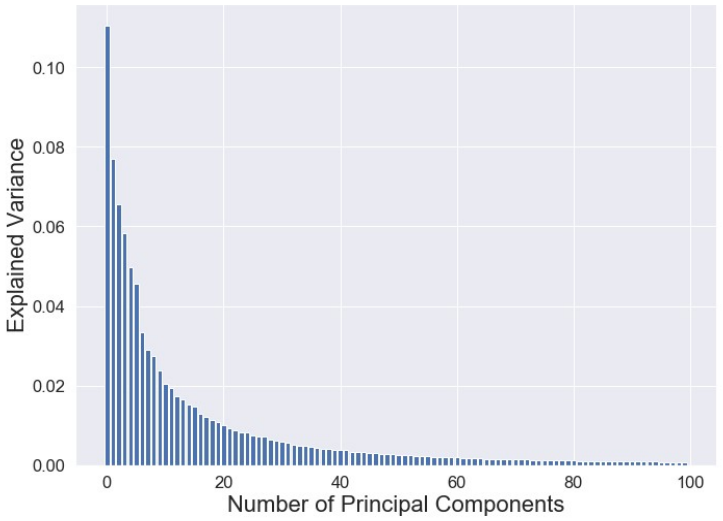
Significant Principal Components

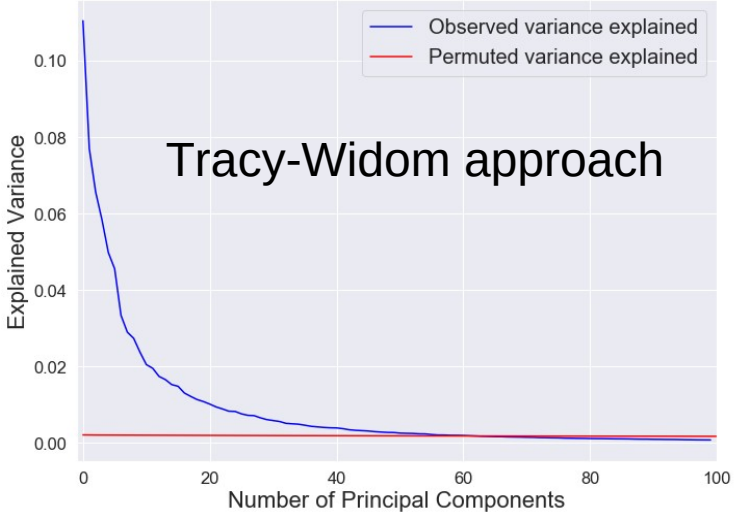
Patterson, Price and Reich: Plos Genetics 2006

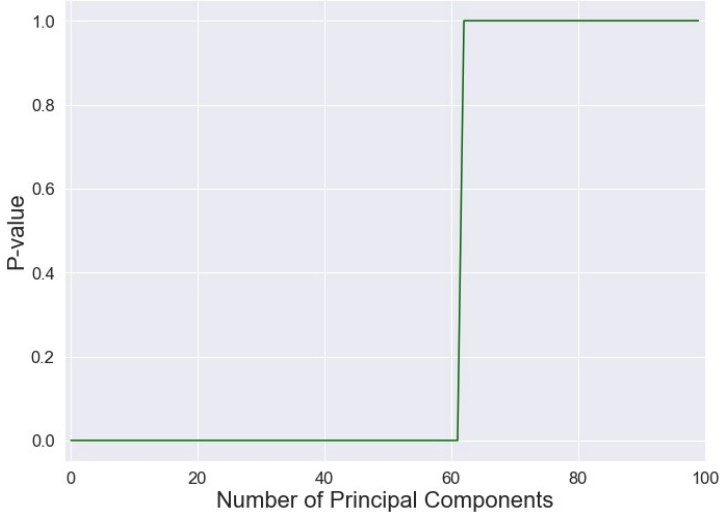
- Largest eigen values follow Tracy-Widom distribution
- Therefore we can implement a formal statistical test for detecting population structure in genetic variation data
- PCA dimensionality reduction is directly related to cluster analysis
- Number of largest PCA eigenvalues is approximately the number of clusters
- There is some sort of “phase transition” when population structure becomes visible

Estimating the number of informative PCs









Statistically Significant Principal Components

COMMENTARY

Check for updates

doi:10.1111/evo.13835

Be careful with your principal components

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Principal components analysis (PCA) is a common method to summarize a larger set of correlated variables into a smaller and more easily interpretable axes of variation. However, the different components need to be distinct from each other to be interpretable otherwise they only represent random directions. This is a fundamental assumption of PCA and, thus, needs to be tested every time. Sample correlation matrices will always result in a pattern of decreasing eigenvalues even if there is no structure. Tests are, therefore, needed to discern real patterns from illusionary ones. Furthermore, the loadings of the vectors need to be larger than expected by random data to be useful in the calculation of PC-scores. PC-scores calculated from nondistinct PC's have very large standard errors and cannot be used for biological interpretations. I give a number of examples to illustrate the potential problems with PCA. Robustness of the PC's increases with increasing sample size but not with the number of traits. I review a few simple test statistics appropriate for testing PC's and use a real-world example to illustrate how this can be done using randomization tests. PCA can be very useful but great care is needed to avoid spurious results.

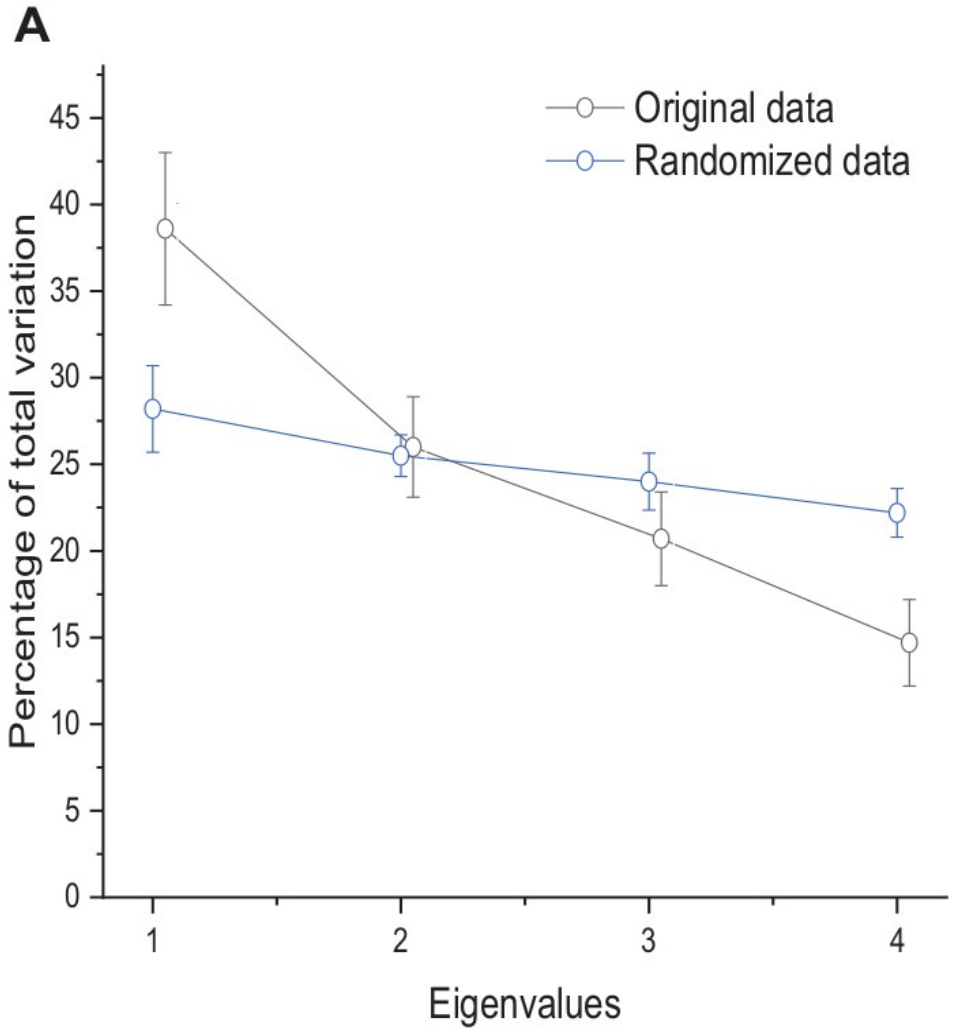
KEY WORDS: Correlations, principal components analysis, randomization, standard error.

A common problem in evolutionary biology is that a researcher has a large number of measurements of various variables that might be interpreted for the understanding of, for example, ecological patterns or multivariate selection. One approach to handle this is to simply test variables one by one. However, these variables can be correlated to each other to various degrees, which means that the tests are not independent and the researcher can drown in a multitude of tests that are not easy to interpret.

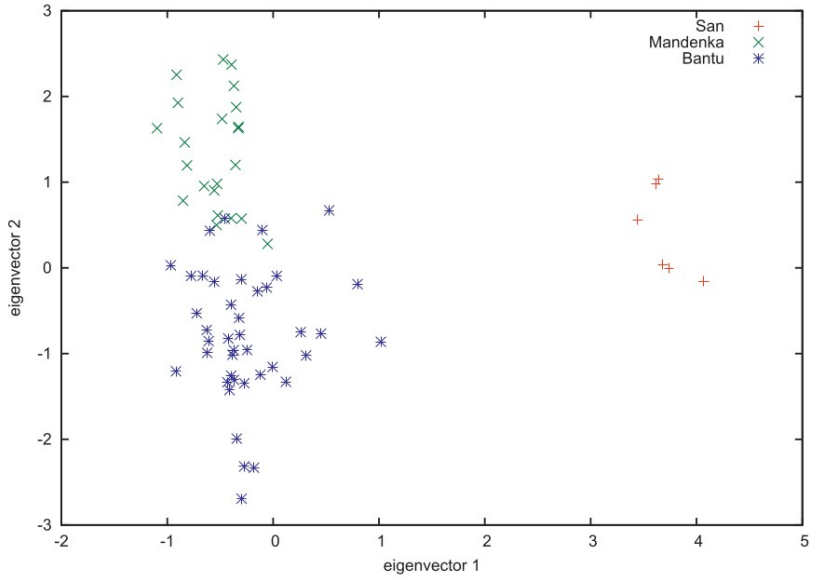
One very common solution is to try to reduce the dimensionality of the data into fewer independent variables that take the interdependence of variables into account, for example, by a principal component analysis (PCA). One example is given in Figure 1 where four variables, let us say morphological measurements, are correlated to various degrees (the number of traits is arbitrarily chosen to for illustration only). The PCA shows that the first principal component (PC) is a size vector (a vector where all loadings are positive and of the products of the loadings and the individual values creating individual PC-scores): In this example this is a size vector and consequently only one correlation is made using the PC-scores. In this case, it can be inferred that size is negatively correlated with the ecological variable. This is a standard procedure to handle complex data in a more easily understood way. This makes a lot of sense and is not a problem per se.

THE PROBLEM

The problem with the example in Figure 1 is that the correlation matrix is a sample correlation matrix based on a sample size of $N = 20$, while the true correlations are all set to 0.0 (for details see Appendix 1). Thus, there is no pattern apart from that created from sampling error alone. Indeed, in all cases where something is estimated, the estimated value will deviate from the true value



Tracy-Widom Significant Axes



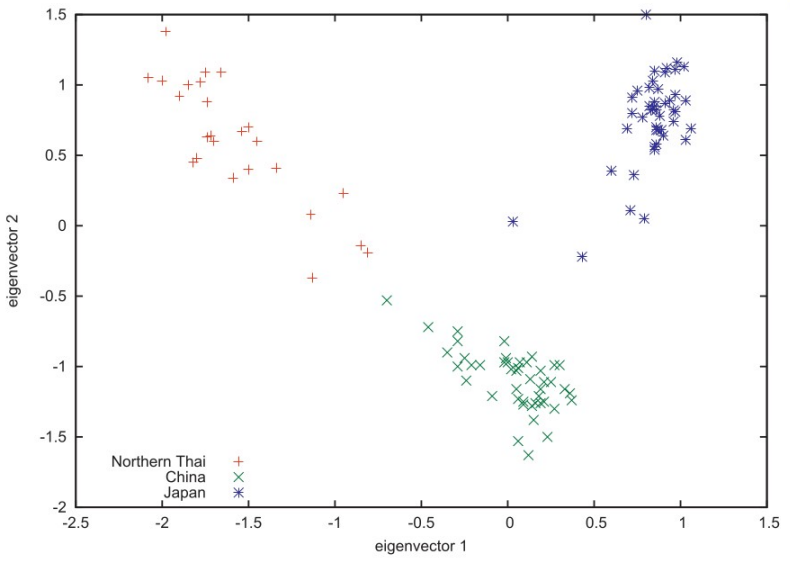


Table 1. Statistics from HGDP African Data

Number	Eigenvalue	TW Statistic	TW *p*-Value	ANOVA *p*-Value
1	2.07	46.2	$<10^{-12}$	$<10^{-12}$
2	1.40	6.717	3.08×10^{-7}	$<10^{-12}$
3	1.31	0.380	.108	.74

in particular samples, populations, and SNPs. For this reason, we only used markers for analysis for which there was no missing data, and we corrected for LD using our regression technique (see below). The details of the data preprocessing

Table 2. Statistics from Thai/Chinese/Japanese Data

Number	Eigenvalue	TW Statistic	TW *p*-Value	ANOVA *p*-Value
1	2.21	92.34	$<10^{-12}$	$<10^{-12}$
2	1.47	31.15	$<10^{-12}$	$<10^{-12}$
3	1.23	−1.61	.61	.97

Table 3. Statistics from Shriver Dataset

Number	Eigenvalue	TW Statistic	TW *p*-Value	ANOVA *p*-Value
1	22.36	76.091	$<10^{-12}$	$<10^{-12}$
2	8.20	106.870	$<10^{-12}$	$<10^{-12}$
3	5.09	106.071	$<10^{-12}$	$<10^{-12}$
4	3.81	103.146	$<10^{-12}$	$<10^{-12}$
5	3.33	115.239	$<10^{-12}$	$<10^{-12}$
6	2.09	60.090	$<10^{-12}$	$<10^{-12}$
7	1.89	51.768	$<10^{-12}$	$<10^{-12}$
8	1.44	14.658	$<10^{-12}$	$<10^{-12}$
9	1.30	2.038	.010	1.09×10^{-7}
10	1.27	0.084	.084	0.78

A third issue is that our methods require that divergence is small, and that allele frequencies are divergent primarily because of drift. We attempted to apply our methods to an African-American dataset genotyped on a panel of ancestry-informative markers [34]. The Tracy–Widom theory breaks down here with dozens of ‘‘significant’’ axes.

Phase Transition in Population Structure

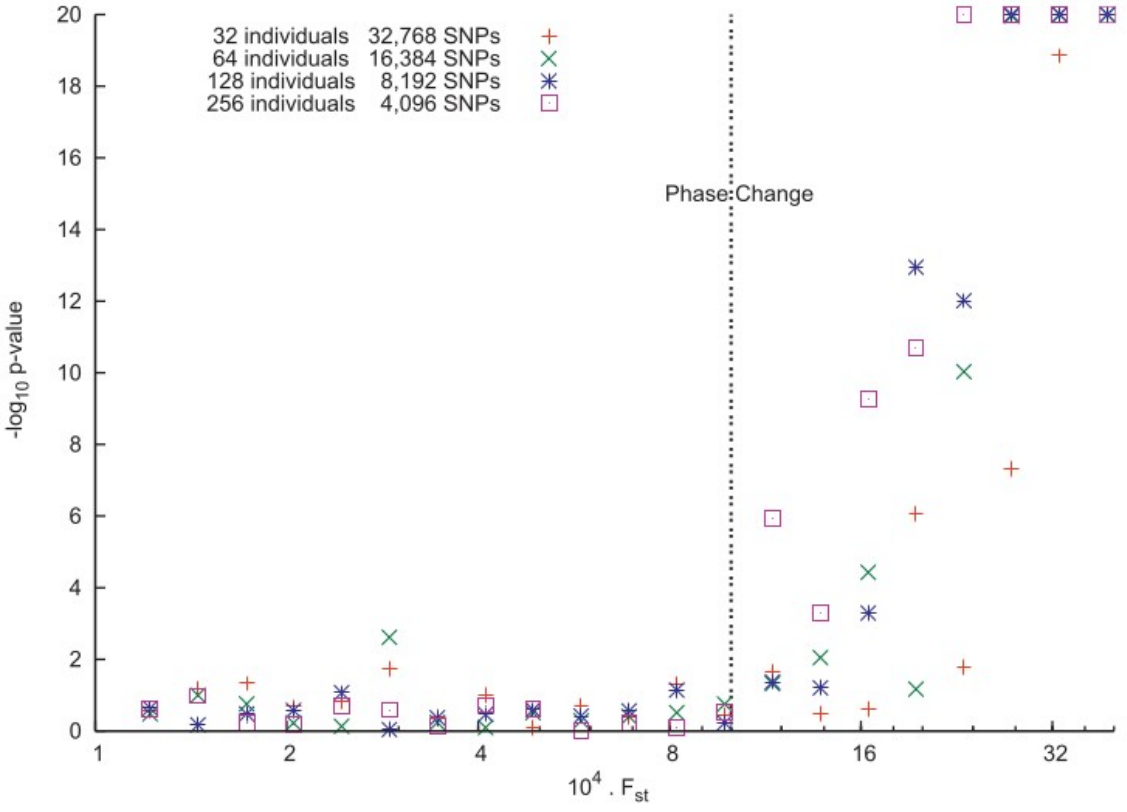


Figure 6. The BBP Phase Change

We ran a series of simulations, varying the sample size m and number of markers n but keeping the product at $mn = 2^{20}$. Thus the predicted phase change threshold is $F_{ST} = 2^{-10}$. We vary F_S and plot the log p-value of the Tracy–Widom statistic. (We clipped $-\log_{10} p$ at 20.) Note that below the threshold there is no statistical significance, while above threshold, we tend to get enormous significance.

doi:10.1371/journal.pgen.0020190.g006

PCA in Population Genetics: Literature Review

Menozzi, Piazza and Cavalli-Sforza

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ARTICLES

Synthetic maps of human gene frequencies in Europeans

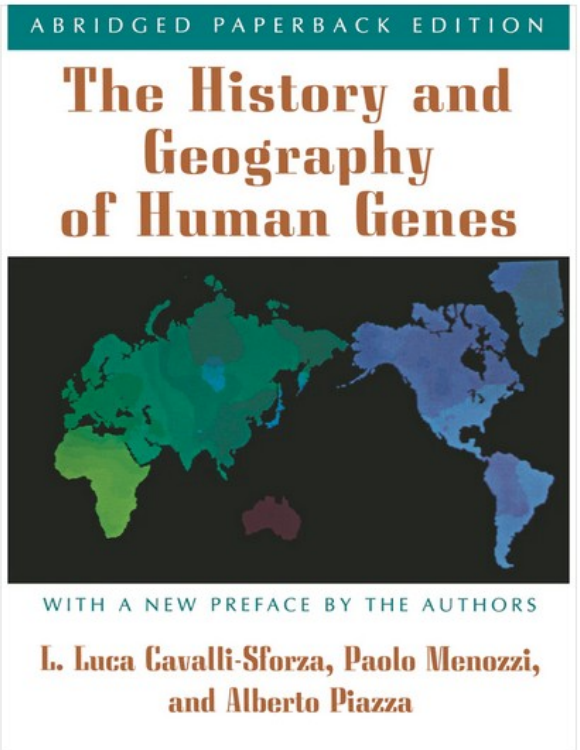
P Menozzi, A Piazza, L Cavalli-Sforza
+ See all authors and affiliations

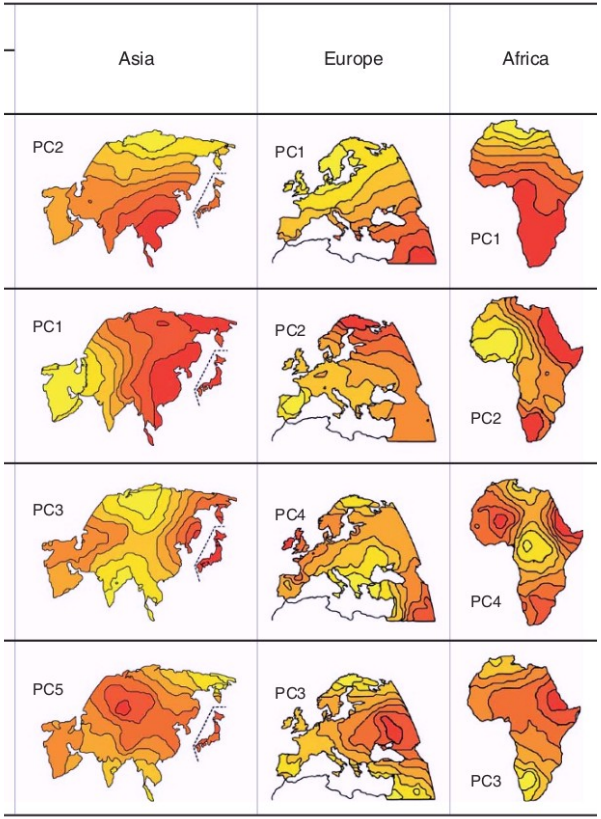
Science 01 Sep 1978:
Vol. 201, Issue 4358, pp. 786-792
DOI: 10.1126/science.356262

Article Info & Metrics eLetters PDF

Abstract

Multivarate techniques can be used to condense the information for a large number of loci and alleles into one or a few synthetic variables. The geographic distribution of synthetic variables can be plotted by the same technique used in mapping the gene frequency of a single allele. Synthetic maps were constructed for Europe and the Near East, with the use of principal components to condense the information of 38 independent alleles from ten loci. The first principal component summarizes close to 30% of the total information and shows gradients. Maps thus constructed show clines in remarkable agreement with those expected on the basis of the spread of early farming in Europe, thus supporting the hypothesis that this spread was a demic spread rather than a cultural diffusion of farming technology.





Genetic distances correlate with geographical distances possibly due to people's migration

Neolithic farming migration rather than a cultural diffusion of farming technology

NEWS AND VIEWS

Principal component analysis of genetic data

David Reich, Alkes L Price & Nick Patterson

Principal component analysis (PCA) has been a useful tool for analysis of genetic data, particularly in studies of human migration. A new study finds evidence that the observed geographic gradients, traditionally thought to represent major historical migrations, may in fact have other interpretations.

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Principal component analysis (PCA) has been used for several decades to study human population migrations, resulting in remarkable inferences about history. On page 646 of this issue, John Novembre and Matthew Stephens[1] show that the geographic gradients that emerge when PCA is applied to genetic data—and that are sometimes interpreted as highly suggestive of major historical migrations—can also have other explanations. We suggest guidelines for scientists interested in using PCA in genetic analysis in light of this potential concern and highlight three applications in which PCA has continued value: detecting population substructure, correcting for stratification in disease studies and making qualified inferences about human history.

Synthetic maps in question

PCA is a statistical method for exploring and making sense of datasets with a large number of measurements (which can be thought of as dimensions) by reducing the dimensions to the few principal components (PCs) that explain the main patterns. Thus, the first PC is the mathematical combination of measurements that accounts for the largest amount of variability in the data. Luca Cavalli-Sforza and colleagues had the original insight that PCA could be applied to human genetic variation[2], and they eventually analyzed

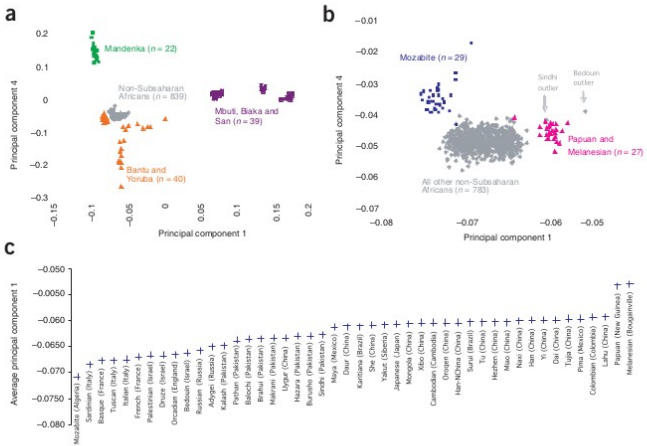


Figure 1 PCA continues to provide evidence of important migration events. (**a**) We carried out PCA on 940 individuals from the Human Genome Diversity Project that were scanned at approximately 650,000 SNPs[11] using data from 101 sub-Saharan African samples to define the PCs (Mandenka, Bantu from Kenya and South Africa, Yoruba, San, Mbuti Pygmy and Biaka Pygmy). We carried out the analysis on samples blinded to population labels (the coloring of samples was only carried out after the analysis). We plotted principal component 1 (negative values are more Bantu-related) and principal component 4 (positive values are more closely related to the Senegalese Mandenka). (**b**) Outlying populations are the Mozabite, who are more Mandenka-related, reflecting recent gene flow across the Sahara, and Papuans and Melanesians, who have inherited less Bantu-related gene flow. (**c**) To reveal the west-to-east gradient of Bantu-related ancestry across Eurasia, we averaged the first PC for each of the non-African populations and plotted the populations in rank order.

LETTERS

nature genetics

Interpreting principal component analyses of spatial population genetic variation

John Novembre[1,3] & Matthew Stephens[1,2]

Nearly 30 years ago, Cavalli-Sforza *et al.* pioneered the use of principal component analysis (PCA) in population genetics and used PCA to produce maps summarizing human genetic variation across continental regions[1]. They interpreted gradient and wave patterns in these maps as signatures of specific migration events[1–3]. These interpretations have been controversial[4–7], but influential[8], and the use of PCA has become widespread in analysis of population genetics data[9–13]. However, the behavior of PCA for genetic data showing continuous spatial variation, such as might exist within human continental groups, has been less well characterized. Here, we find that gradients and waves observed in Cavalli-Sforza *et al.*'s maps resemble sinusoidal mathematical artifacts that arise generally when PCA is applied to spatial data, implying that the patterns do not necessarily reflect specific migration events. Our findings aid interpretation of PCA results and suggest how PCA can help correct for continuous population structure in association studies.

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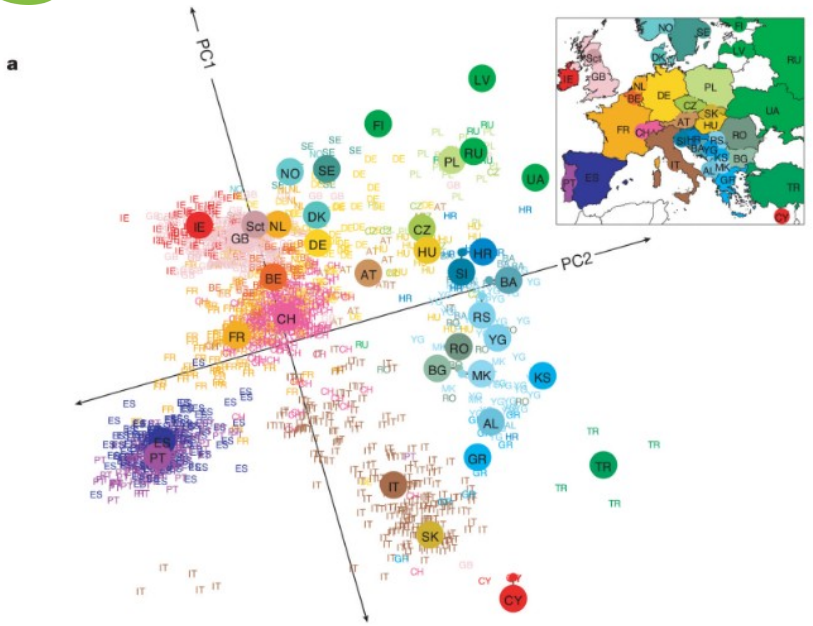
Cavalli-Sforza *et al*'s classic text "The History and Geography of Human Genes"[3] synthesizes a decades-long survey of human genetic variation. These ground-breaking datasets stimulated development of methods that are now widely used, including application of principal component analysis (PCA) to population genetic variation. In essence, Cavalli-Sforza *et al.* collected count data for many genetic variants

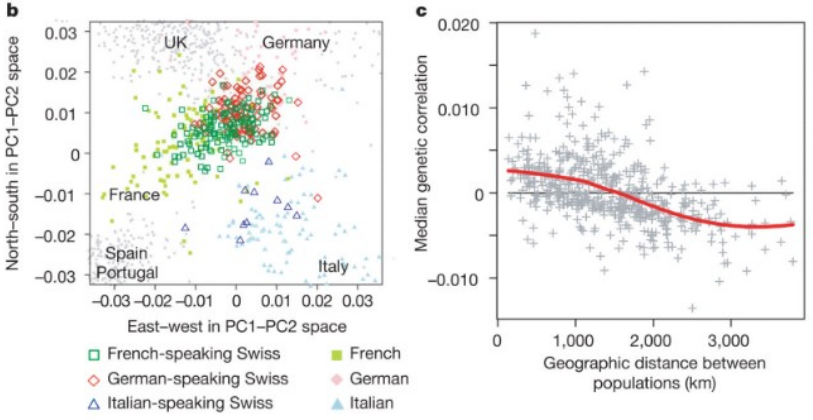
origin must be the center of the radiation" (p. 295 of ref. 3). They also suggest centripetal population movements as an alternative explanation. Examples of their explanations for the European PC maps in **Figure 1** include expansion of agriculturalists out of the Near East (Europe PC1); migrations of Mongoloid Uralic speakers from northwestern Asia (Europe PC2); migration of the carriers of the proto-Indo-European Kurgan culture in Europe (Europe PC3); and an expansion from Greece (Europe PC4).

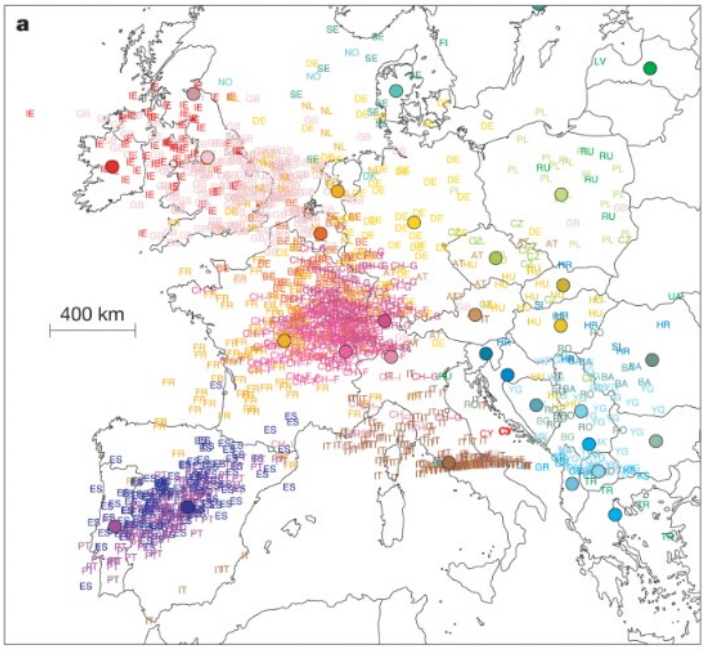
Because the basis for these interpretive guidelines is unclear, we performed simulations to investigate whether such specific migration events are necessary to explain the observed patterns. Specifically, we performed PCA on data simulated under equilibrium population genetic models without range expansions, assuming a constant homogeneous short-range migration process across both time and (two-dimensional) space. The results showed highly distinctive structure. For example, the first two PC maps show large-scale orthogonal gradients, and the next two show 'saddle' and 'mound' patterns (**Fig. 1**). The same four basic patterns occurred consistently in the first few PC maps across multiple simulations, although not always in the same order (**Supplementary Fig. 1** online). Results for the analogous one-dimensional habitat setting are even more structured, resembling sinusoidal functions of increasing frequency (**Fig. 2**, **Supplementary Fig. 2** online). Thus PC maps show local peaks and troughs *even when underlying migration patterns are homogeneous across time and space*. This suggests that the local features of the PC

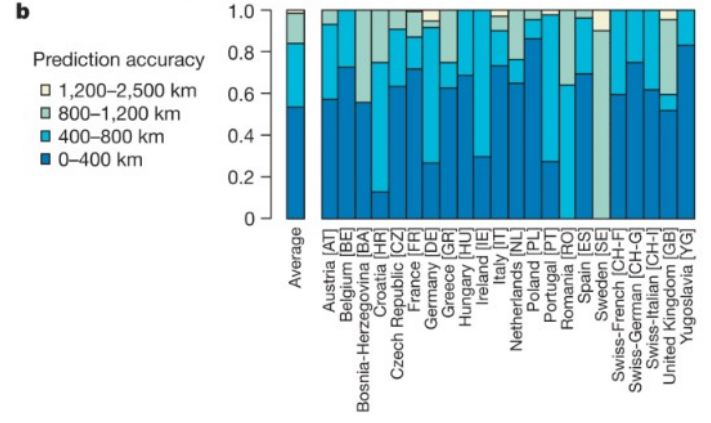
Genetic distances correlate with geographical distances possibly due to isolation by distance and not necessarily migration

More from John Novembre, Nature 2008

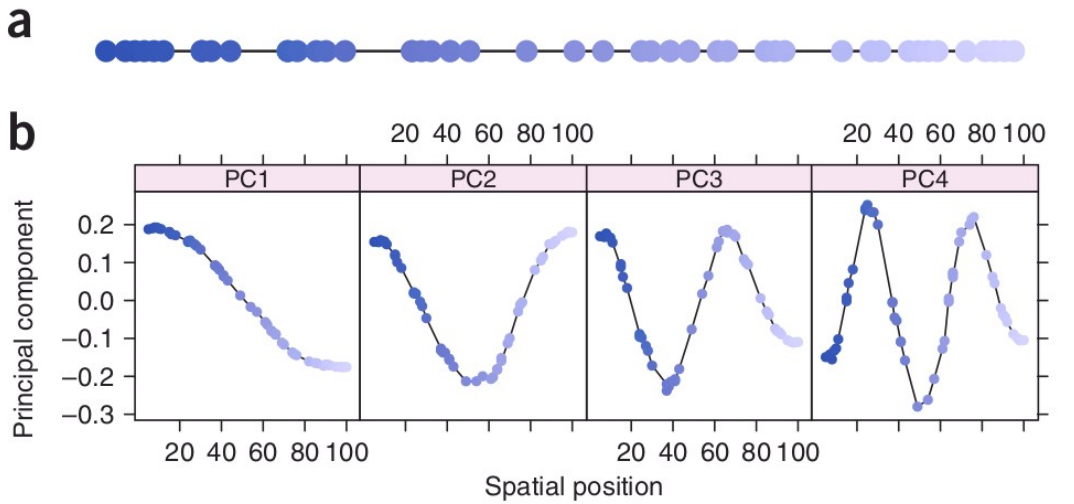




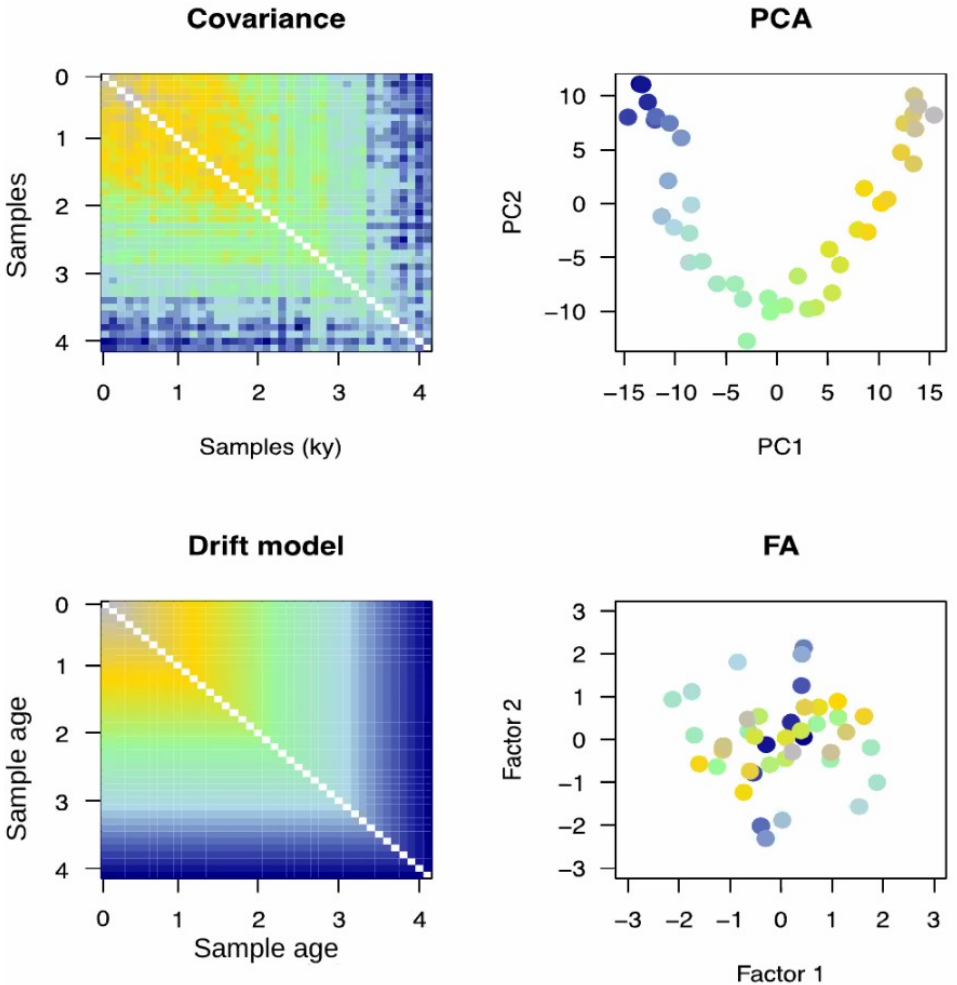




Horseshoe Effect: Spatial and Temporal

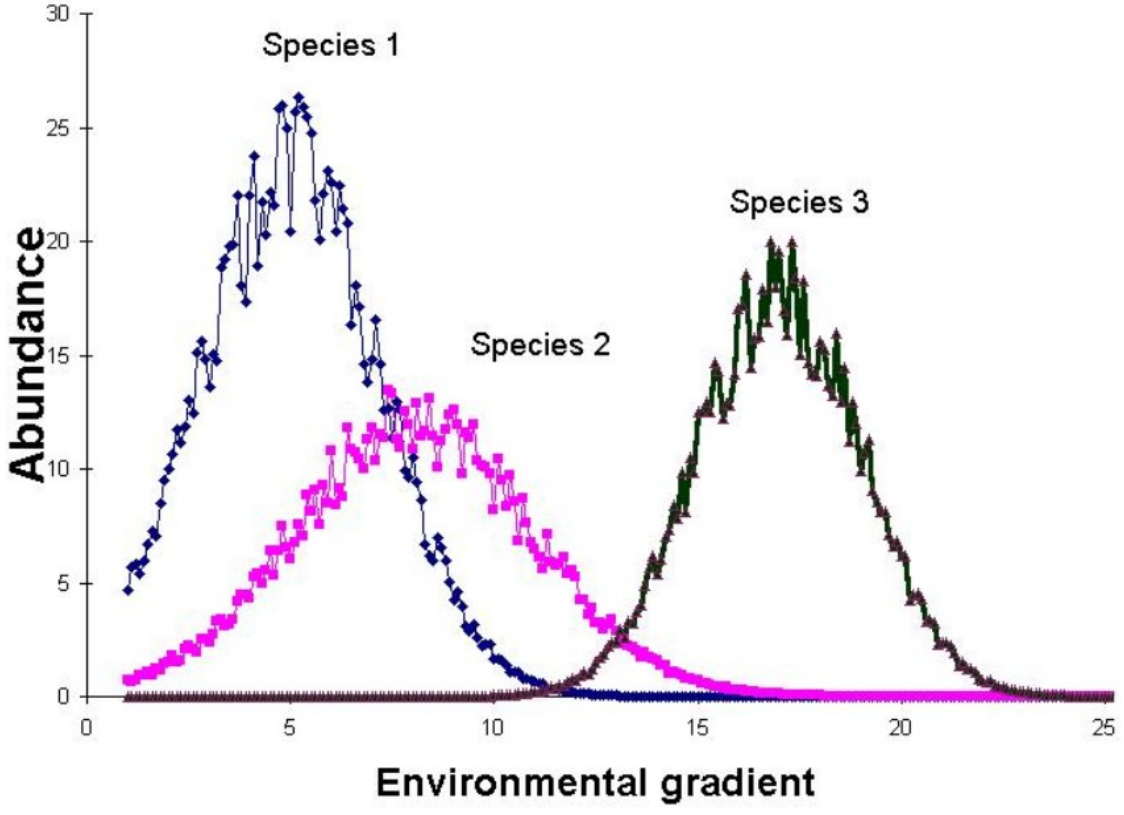


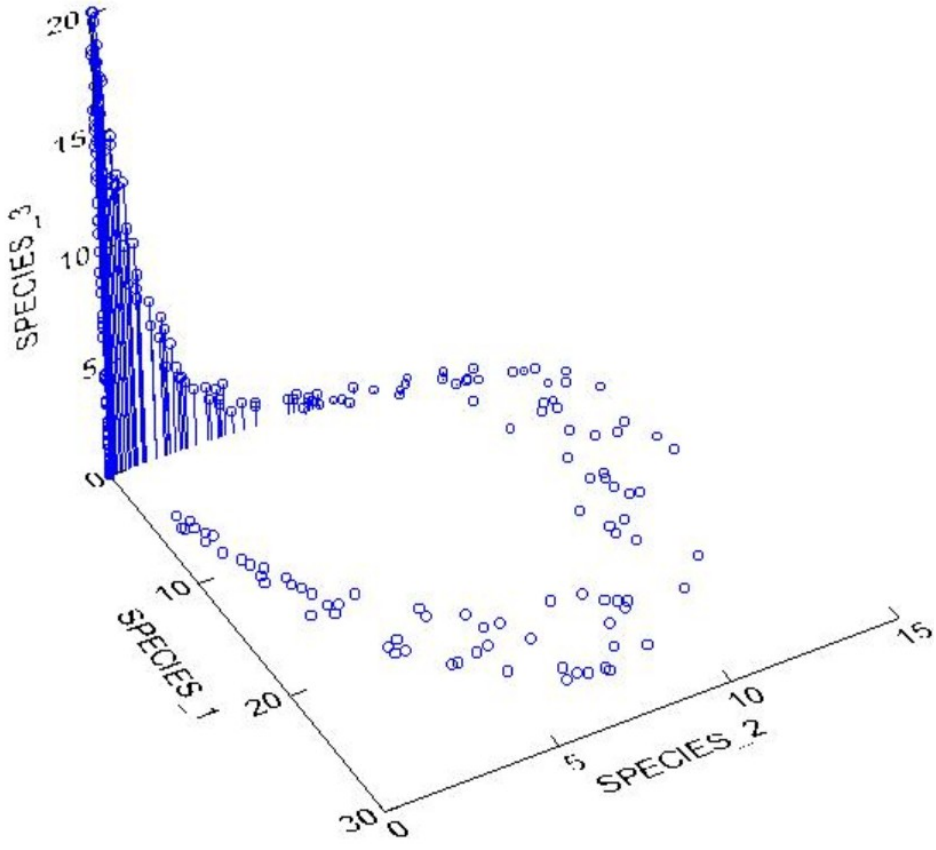
Novembre & Stephens, Nat. Genetics 2008



Francois & Jay, Nat. Communications 2020

Horseshoe Effect: Ecology Example





PCA: Effect of Uneven Sampling

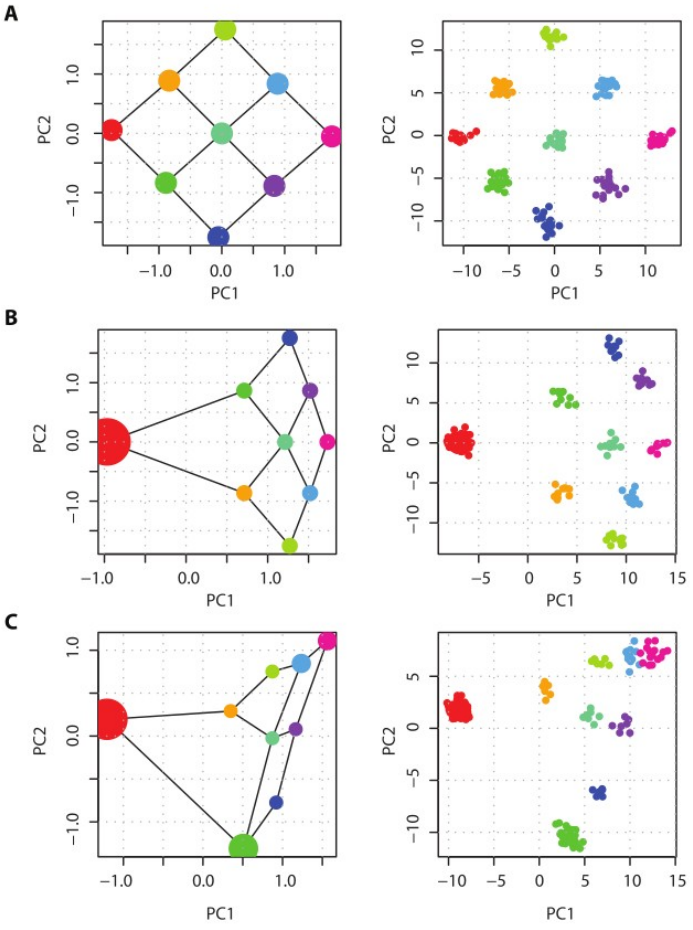


Figure 3. The effect of uneven sampling on PCA projection. PCA projection of samples taken from a set of nine populations arranged in a lattice, each of which exchanges migrants at rate M per N_e generations with each adjoining neighbour, leads to a recovery of the migration-space if samples are of equal size (A), or a distortion of migration-space if populations are not equally represented (B,C). In each part the left-hand panel shows the analytical solution (the area of each point represents the relative sample size) with migration routes illustrated while the right-hand panel shows the result of a simulation with a total sample size of 180 and 10,000 independent SNP loci. All examples are for $M = 2$.
doi:10.1371/journal.pgen.1000686.g003

"The results described here provide an explanation. First, from Equation 10 it can be seen that the the matrix M is influenced by the relative sample size from each population through the components t_i . For instance, even if all populations are equally divergent from each other, those for which there are fewer samples will have larger values of t_i because relatively more pairwise comparisons are between populations"

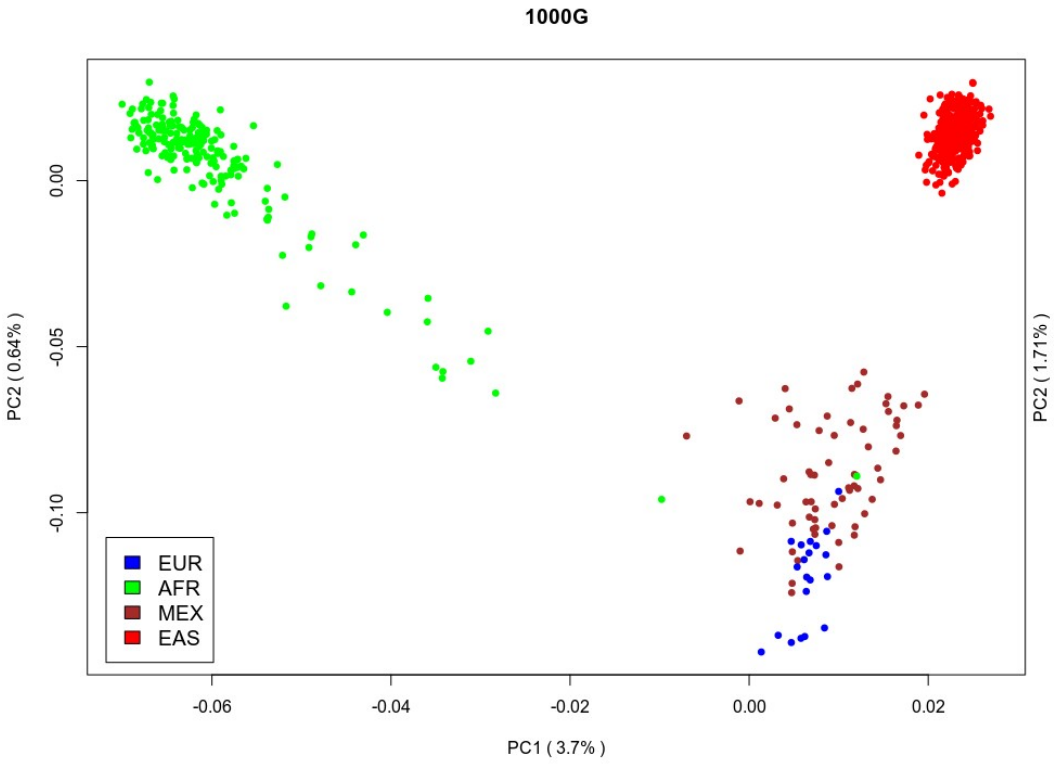
$$A = XX^T = \frac{1}{N}\sum_{ij} x_i x_j$$

$$N = N_{pop1} + N_{pop2} + N_{pop3} + \ldots = \sum_k N_k$$

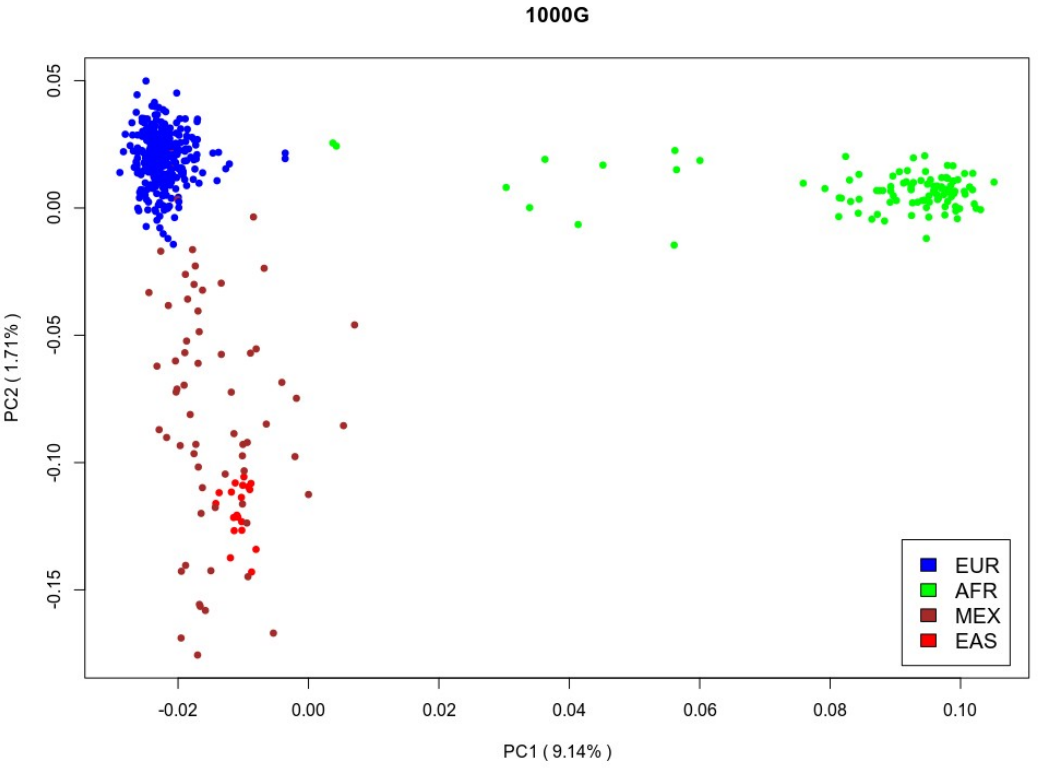
$$A_{uneven} = \sum_{ijk} \frac{1}{N_k} x_{ik} x_{jk}$$

McVean G., PLoS Genet 5(10), 2009

1000G: Uneven Sampling of Populations

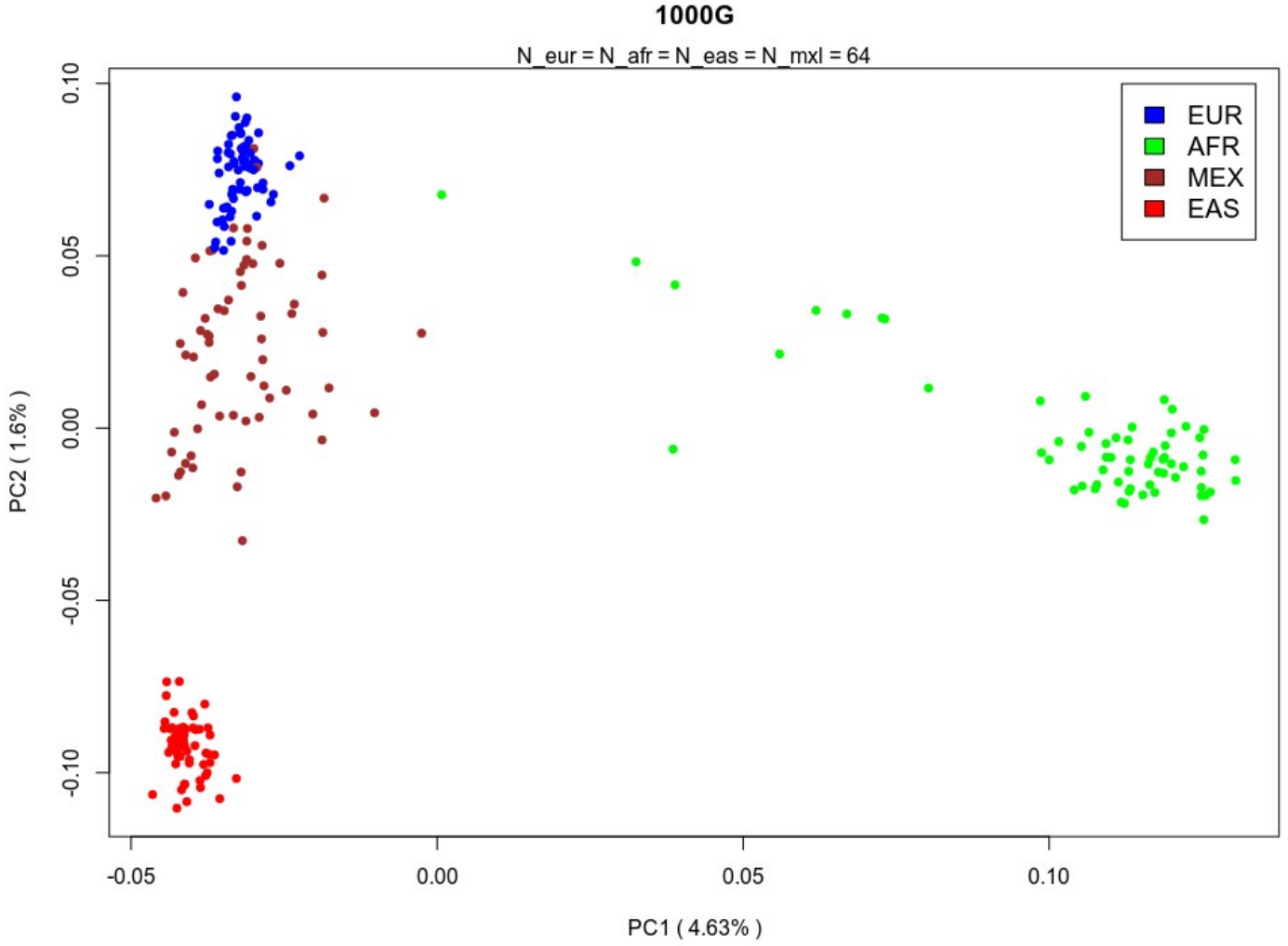


Downsampled Europeans



Downsampled Asians

1000G: Even Sampling of Populations



1000G: Correcting for Uneven Sampling

Proceedings, 10th World Congress of Genetics Applied to Livestock Production

Correcting For Unequal Sampling in Principal Component Analysis of Genetic Data

W. Burgos-Paz[1,2], **S. E. Ramos-Onsins**[1], **M. Pérez-Enciso**[1,2,3], **L. Ferretti**[4]

[1] Centre for Research in Agricultural Genomics (CRAG), 08193 Bellaterra, Spain, [2] Departament de Ciència Animal i dels Aliments, Universitat Autònoma de Barcelona, 08193, Bellaterra, Spain, [3] Institut Català de Recerca i Estudis Avançats (ICREA), Carrer de Lluís Companys 23, Barcelona, 08010, Spain, [4] UMR 7138, UPMC and CIRB, College de France, Paris, France

ABSTRACT: Principal component analysis (PCA) is one of the most widely used tools to explore variability of high dimensional data. PCA is used for population and quantitative genetics. Its popularity has recently increased due to the huge amount of molecular markers available in datasets worldwide. In genetics, a common issue due to external constraints is uneven sampling of populations, limiting the usefulness of PCA because of well-known sample size sensitivity and two-dimensional projection bias. Here we evaluated the use of weighted PCA (wPCA) in genetic data in order to correct uneven sampling bias. Simulations suggest that wPCA improves the two-dimensional projections of PCA data and, in some cases, recovers population relationships patterns, even when sample size is as low as n=1. We used this correction in pig data from populations with uneven sampling, recovering a more realistic structure than inferred with only PCA.

Keywords: SNP; population structure; phylogeography

Introduction

The Principal Component Analysis (PCA) is a widely used tool for visualization of the structure of a highly multidimensional data set. It has applications in many fields, including genetics (Novembre and Stephens (2008)). The idea behind this technique is to reduce the complexity of the data by retaining only the linear combinations of variables explaining the maximum variance in the data set. Individual data is then projected onto the space of these combinations and visualized as a low-dimensional plot. In population genetics, PCA is frequently applied to visualize the genetic structure of populations, and its popularity has increased in the last years because of the high throughput genotyping technologies (e.g., SNP arrays), where it is necessary to extract the underlying structure of the dataset manner (Paschou et al., (2007)).

Despite several advantages of PCA in the population genetics field, this technique is very sensitive to the choice of the dataset and the distortion of the PCA plots due to a biased or unequal sampling is a known problem (McVean (2009)). In genetic studies, it is often not possible to sample individuals according to some criteria chosen a priori. For instance, the choice of sampling could be biased by incomplete knowledge of the population structure, conservation or there could also be ethical issues. Further, many factors, like budget constraints or accessibility of geographical regions, availability of samples and technical problems in their conservation and sequencing limit sampling in the field.

McVean (2009) suggested to correct for unequal sampling by downsampling the different populations, but this may lead to a loss in power or in a less accurate picture of genetic structure. A natural framework for this is to correct for this issue by means of a weighted PCA (wPCA), where the variables measured for each sample can be assigned a different weight w_i (Kriegel et al., (2008)). Kriegel et al. proposed this correction for synthetic data to evaluate the influence of outliers on PCA but, to our knowledge, its properties in genetic data are unknown. Here, we evaluated the usefulness of this correction with data from SNPs polymorphism under different population relationships, and illustrate the method in pig populations where differences in sample size could under-over estimate population relationships.

Materials and Methods

Weighted PCA. Given a set of genetic data of n sequences/genotypes and s biallelic SNPs for each sequence, the first step of the PCA is to compute the covariance:

$$Cov\left(x_s, x_{s'}\right) = E\left(x_s, x_{s'}\right) - E\left(x_s\right)E\left(x_{s'}\right)$$

$$= \frac{1}{n}\sum_{ij} x_s^i x_{s'}^j - \left(\frac{1}{n}\sum_i x_s^i\right)\left(\frac{1}{n}\sum_j x_{s'}^j\right)$$

A common assumption is that all individuals should be equally important in the PCA (that is, the implicit weight of each individual is $1/n$). This is usually correct, since there is often no prior information about the origin and grouping of individuals. However, this is not true for samples taken from different breeds or structured natural populations living in different habitats or geographical locations, since the degree of relationship will differ

In the wPCA paradigm, each sample's genotype can be assigned a different weight w_i; then, the covariance can be rewritten naturally as:

$$Cov\left(x_s, x_{s'}\right) = \sum_i w_i x_s^i x_{s'}^i - \left(\sum_i w_i x_s^i\right)\left(\sum_j w_j x_{s'}^j\right),$$

where the weights are numbers between $0 < w_i < 1$ with $\sum_i w_i = 1$. This reduces to the usual PCA when $w_i = 1/n$. We denote the number of populations by ***npop*** and the number of sequences in population A by nA. Since the weight of population A is the sum of the weights of all the sequences belonging to A, a natural weight for each sequence of A is:

$$w_i = \frac{1}{n_{pop}} \cdot \frac{1}{n_A}, \qquad i \in A$$

Simulations. To evaluate the usefulness of wPCA in genetic data, we simulated six populations considering two scenarios where population relationships are mainly shaped by the individuals exchange (e.g., migrations or introgression). In the first scenario (M1), we considered that all populations are equally related in terms of exchange of individuals. This scenario may occur in cooperative breeding schemes where sires or dams are regularly interchanged between herds. In the second scenario (M2), a subpopulation structure was considered where some populations have high individual exchange rates among them but low with the rest. This will typically occur, e.g., when one nucleus provides genetic material to other herds that are otherwise not connected.

For each population, five hundred individuals with thousand independent SNPs each were simulated by coalescence using MLCOASIM.v2 (available at http://bioinformatics.cragenomica.es/numgenomics/people/sebas/software/software.html). Effective population size (Ne=1000) was similar for all populations. Further, each population was sampled retaining one hundred individuals each for downstream analyses. To simulate unbalanced

the PCA projection with an unbalanced finite dataset using the usual (middle) and weighted (right) approach. An unweighted PCA result in highly distorted PC projections, in agreement with McVean, (2009). The PCA distortion was especially noticeable in M1, where individuals from reduced populations were located in the middle of the populations with larger sample size. In M2, distortion may cause misinterpretation of subpopulation structure (i.e blue dot individuals).

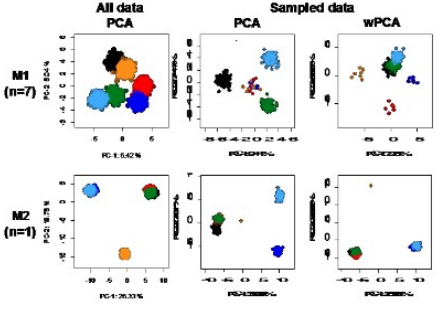


Figure 1. PCA and wPCA projections of simulated

1000G: Correcting for Uneven Sampling

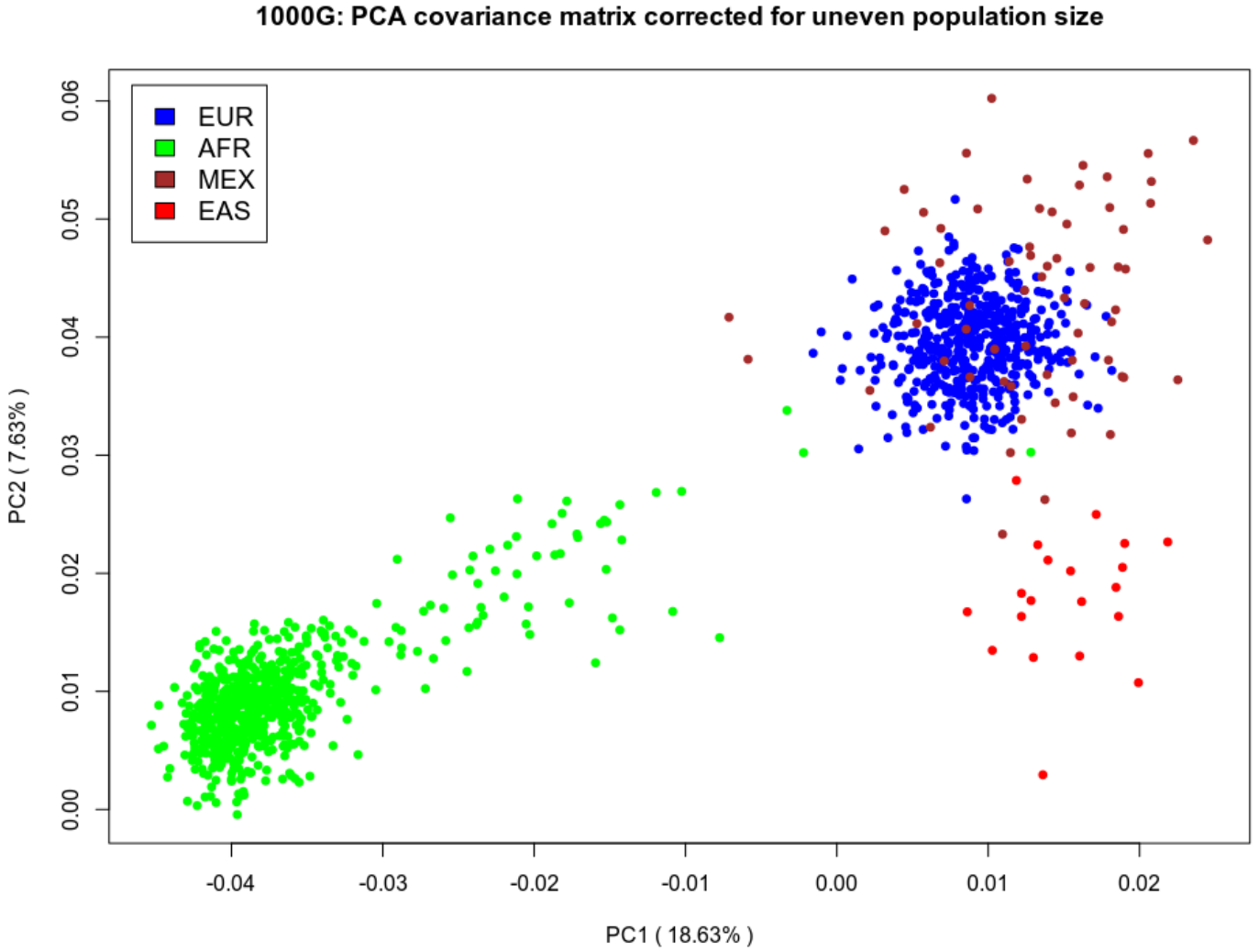
$$A = XX^T = \frac{1}{N}\sum_{ij} x_i x_j$$

$$N = N_{pop1} + N_{pop2} + N_{pop3} + \ldots = \sum_k N_k$$

$$A_{uneven} = \sum_{ijk} \frac{1}{N_k} x_{ik} x_{jk}$$

Normalize each sample by its population size before computing the covariance matrix

Is it still an unsupervised technique?



National Bioinformatics Infrastricture Sweden (NBIS)



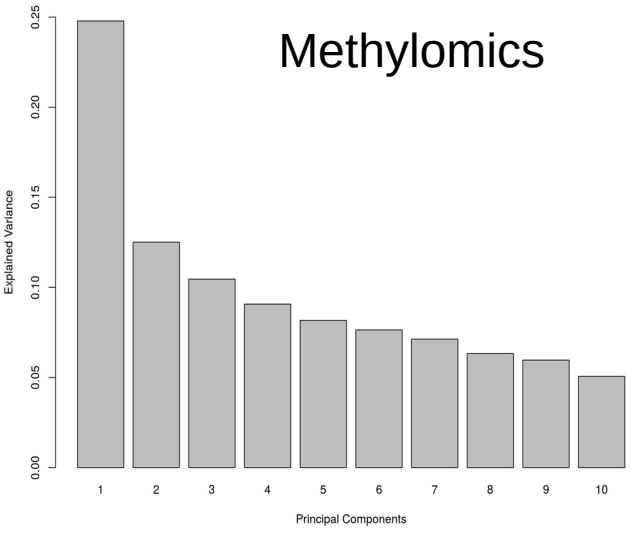
Knut och Alice Wallenbergs Stiftelse

Vetenskapsrådet

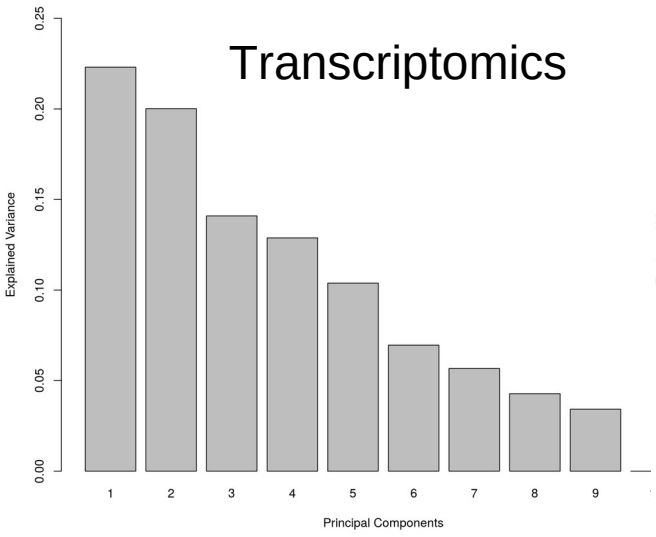
LUNDS UNIVERSITET

Variance Explained by PCs Across OMICs

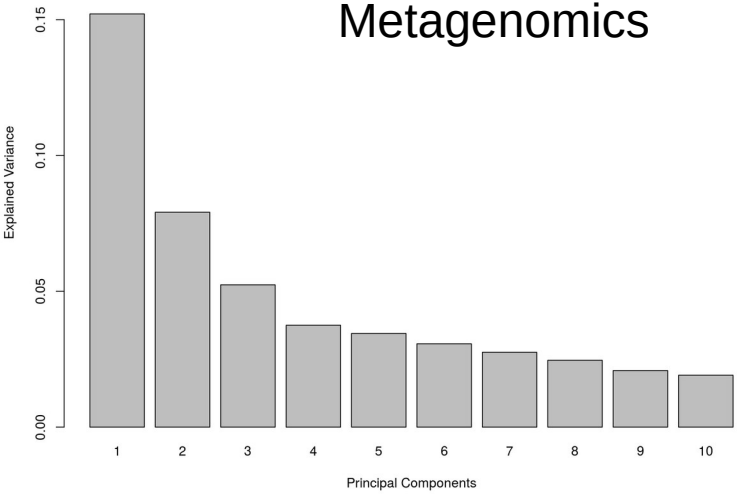
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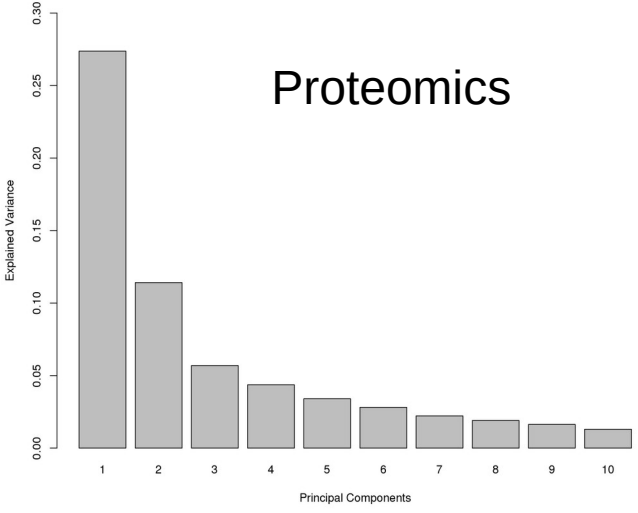
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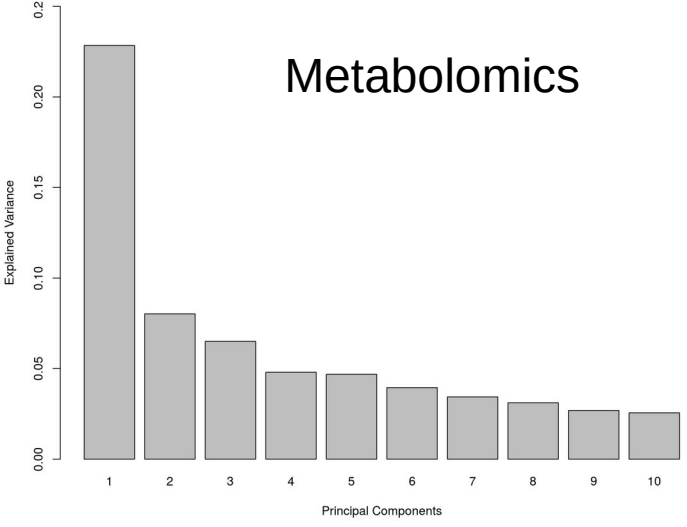
Metagenomics



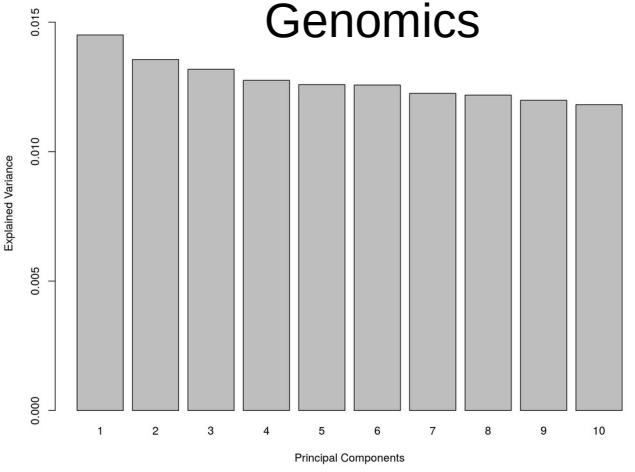
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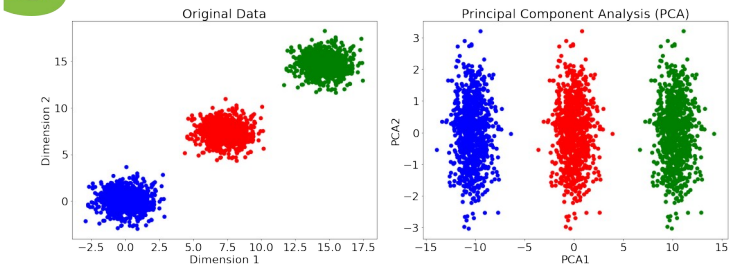
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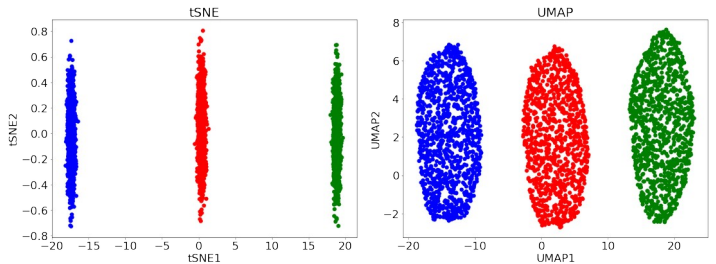


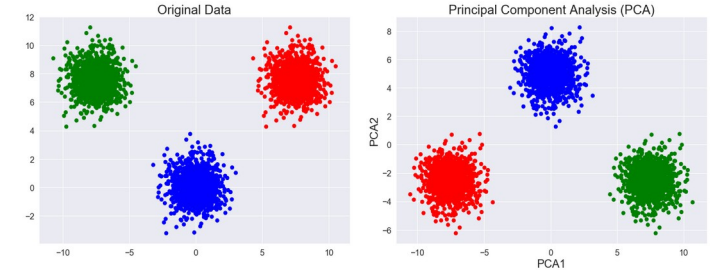
Genomics

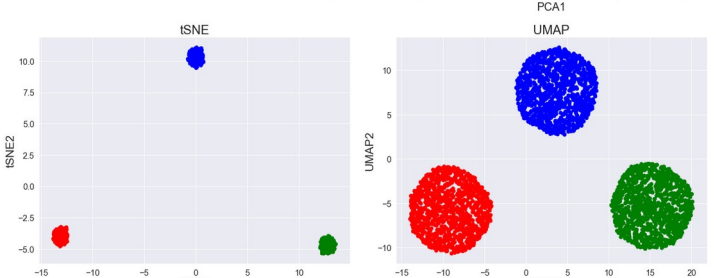


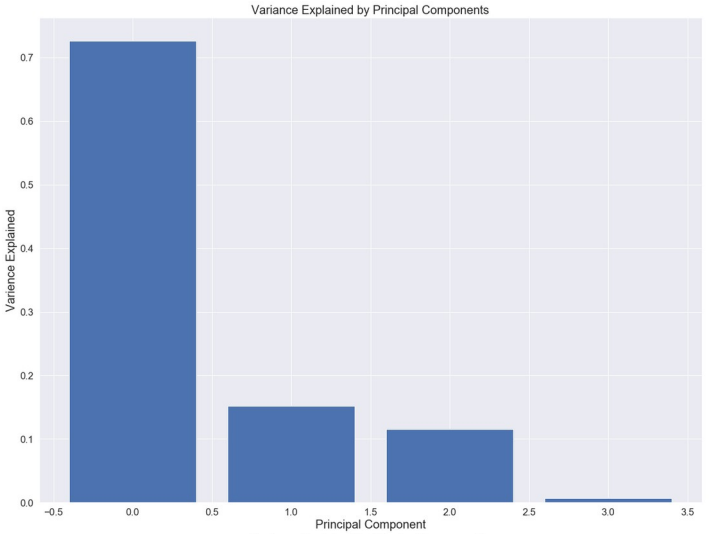
Symmetry of scRNAseq vs Genetic Variation



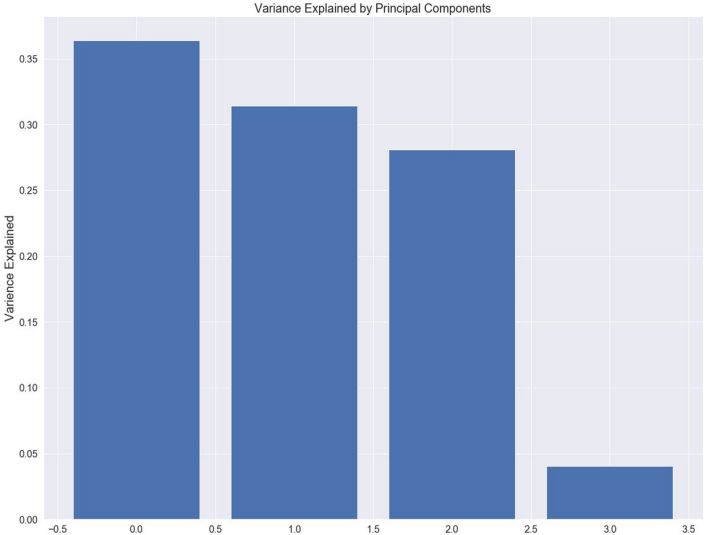






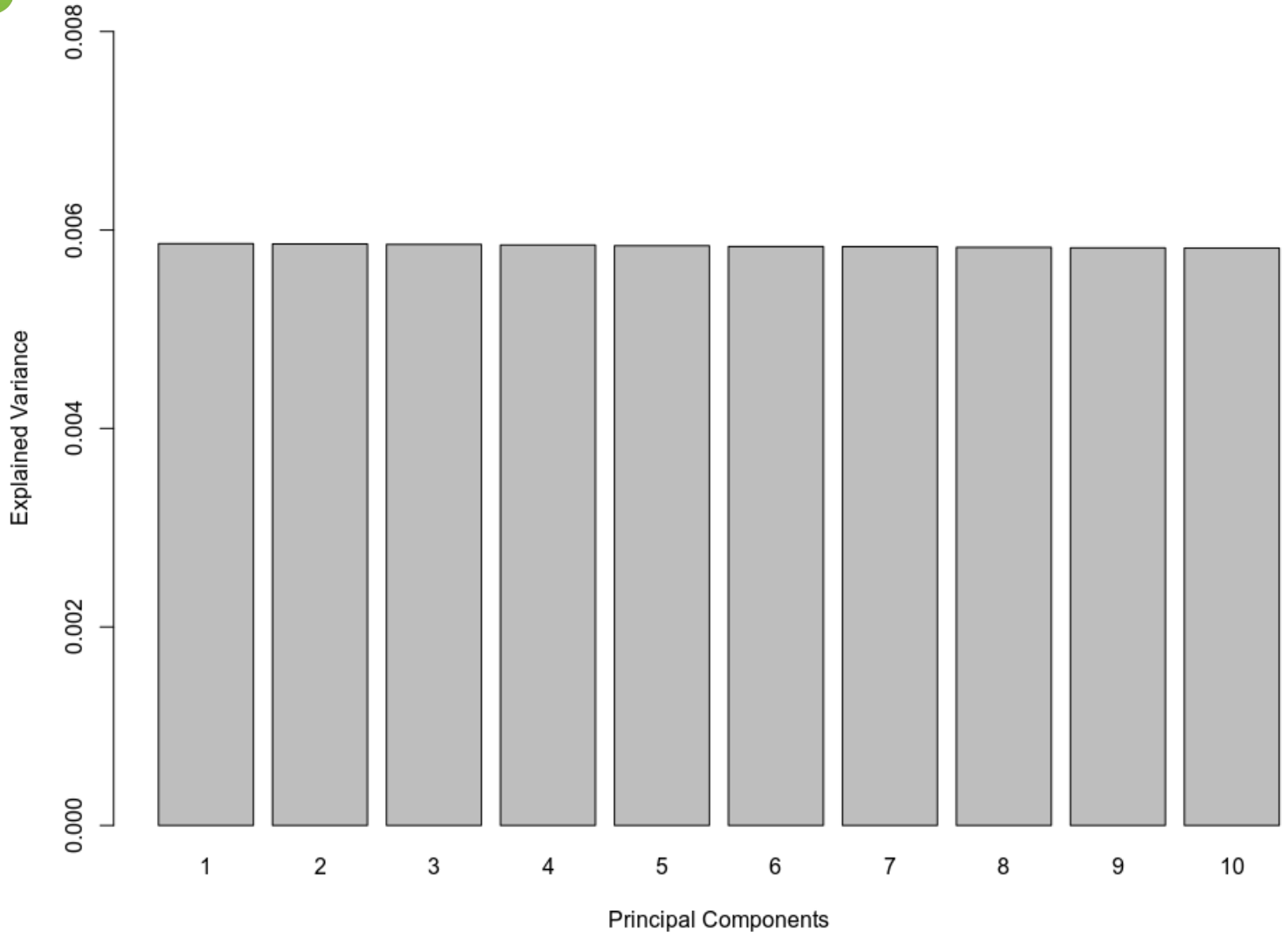


scRNAseq

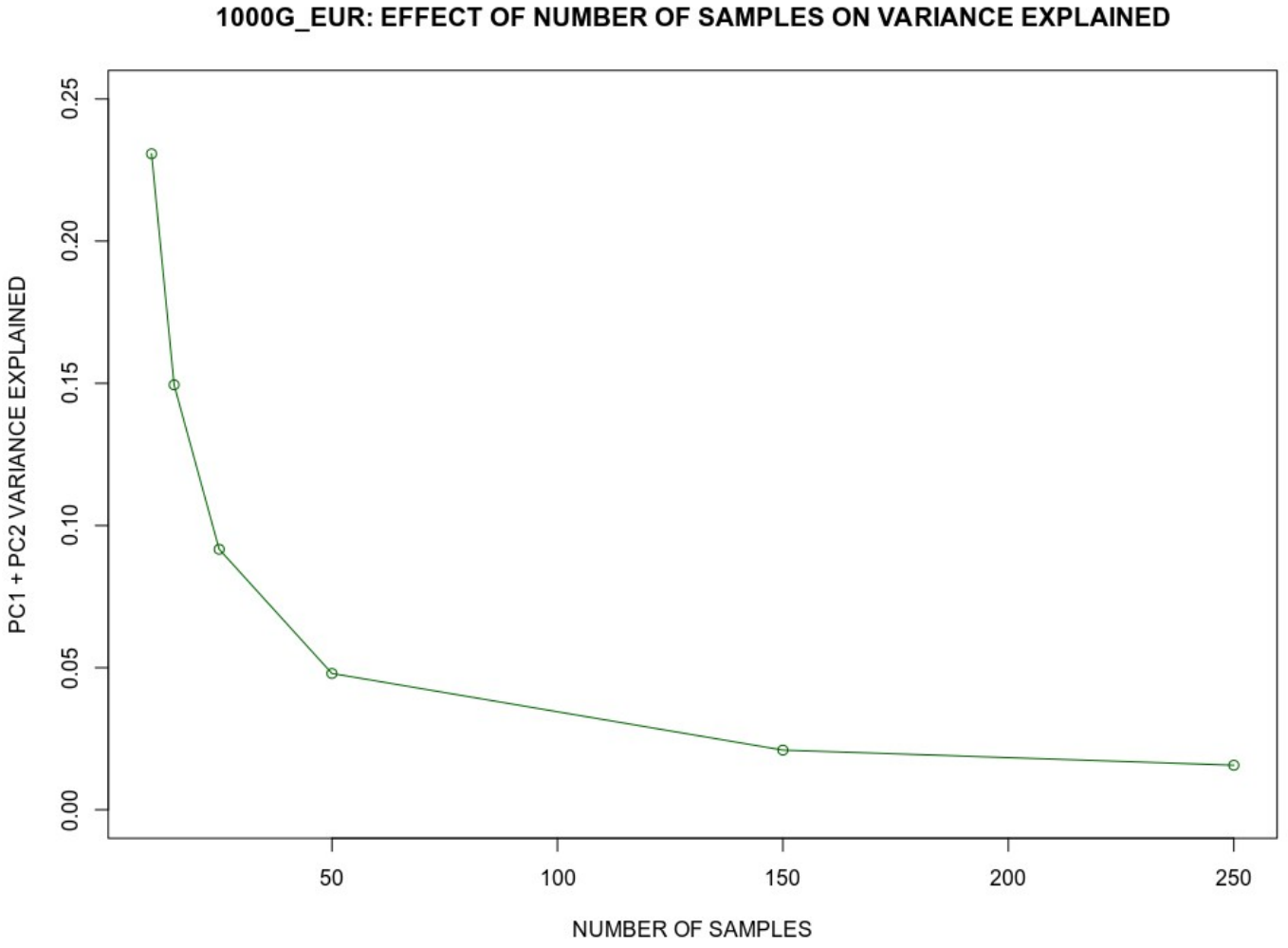


Genetic Variation

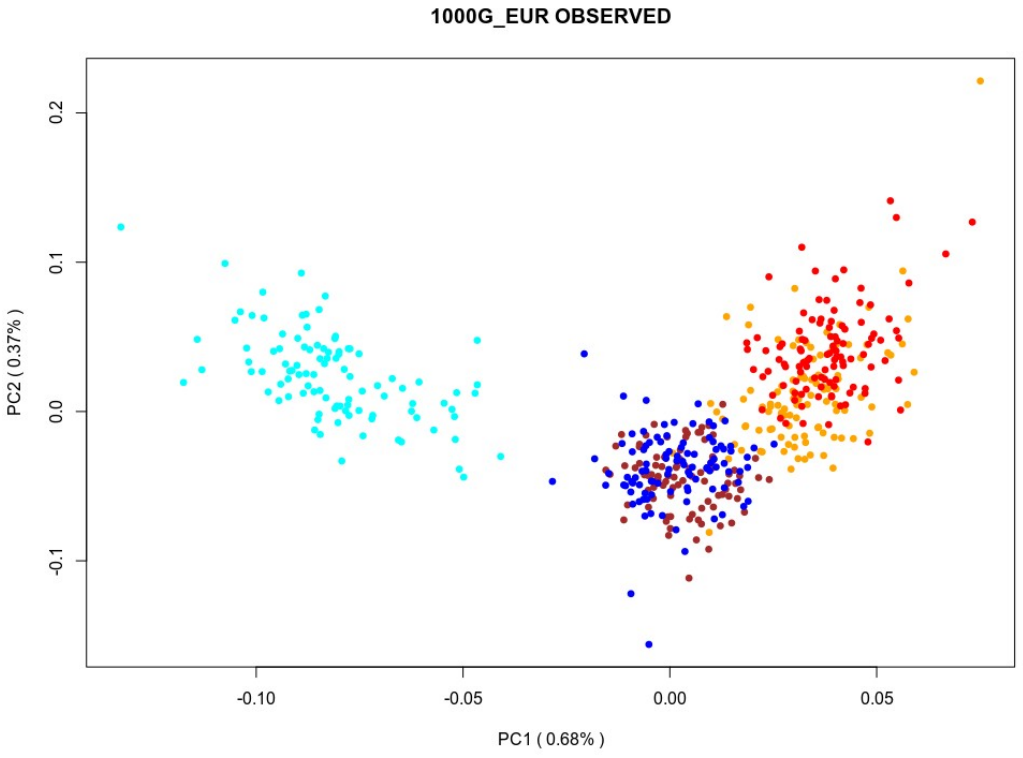
Lack of Variation in Genetic Variation: GWAS

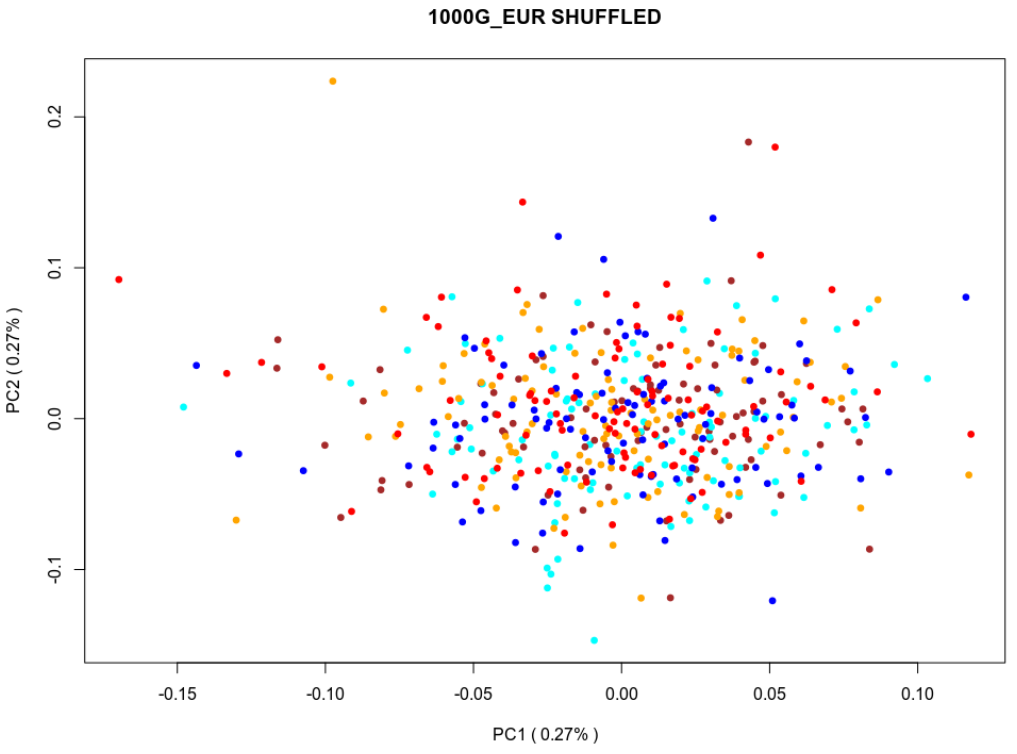


Lack of Genetic Variation or Sample Size?

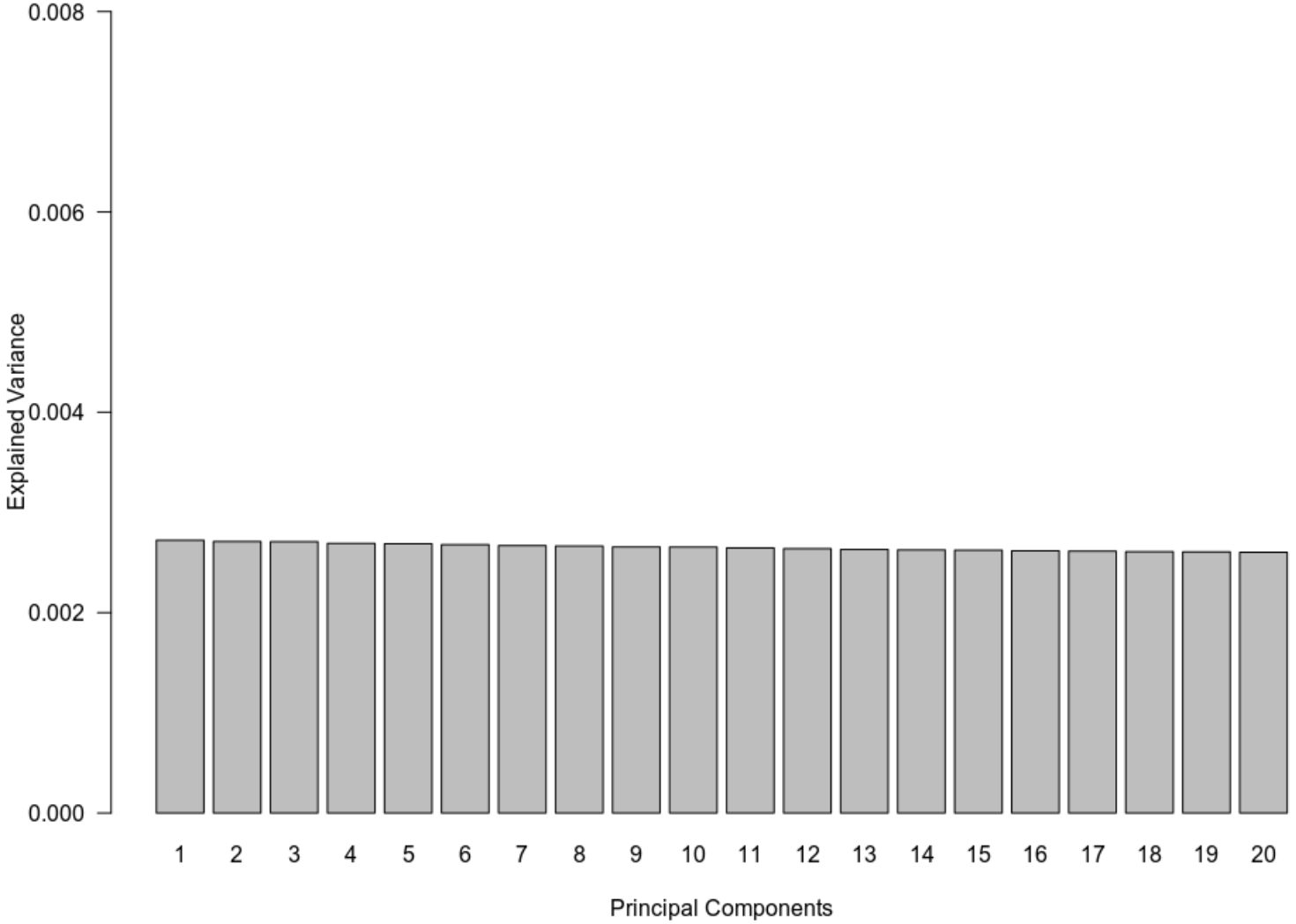


Randomization in 1000G_EUR Data

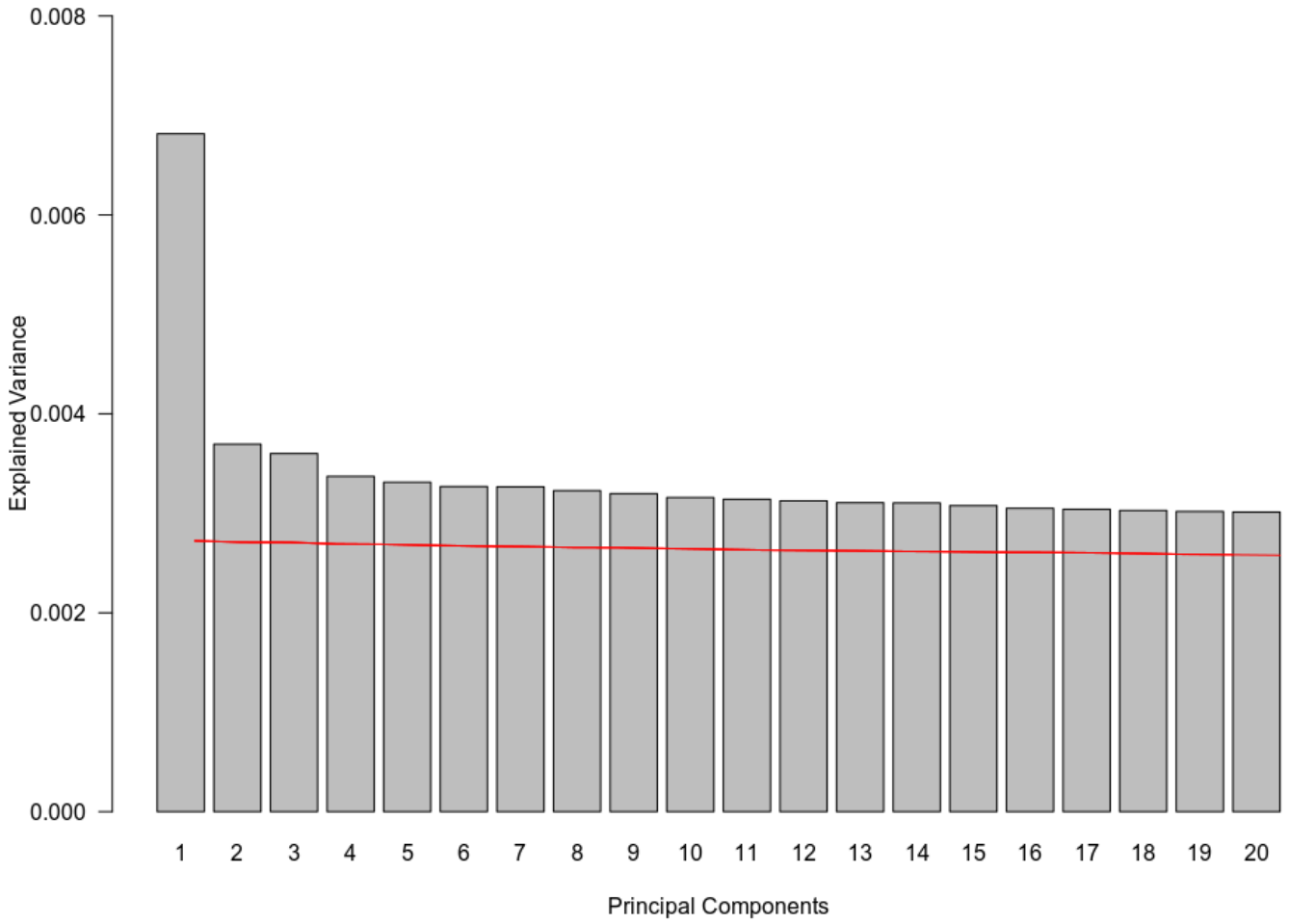




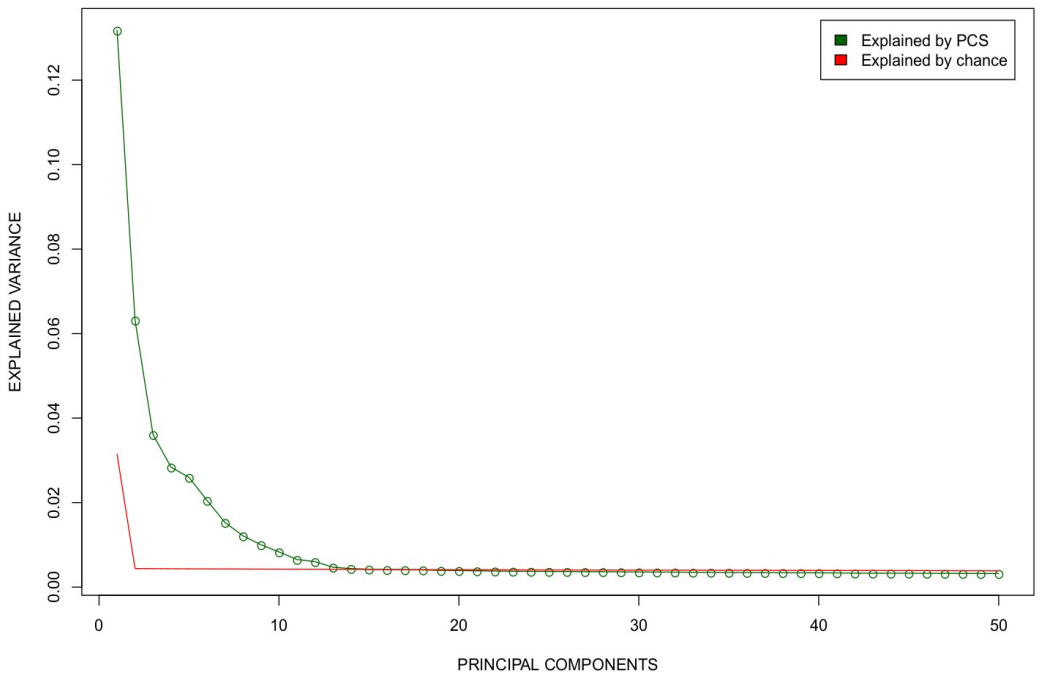
Low Variation in 1000G_EUR Data

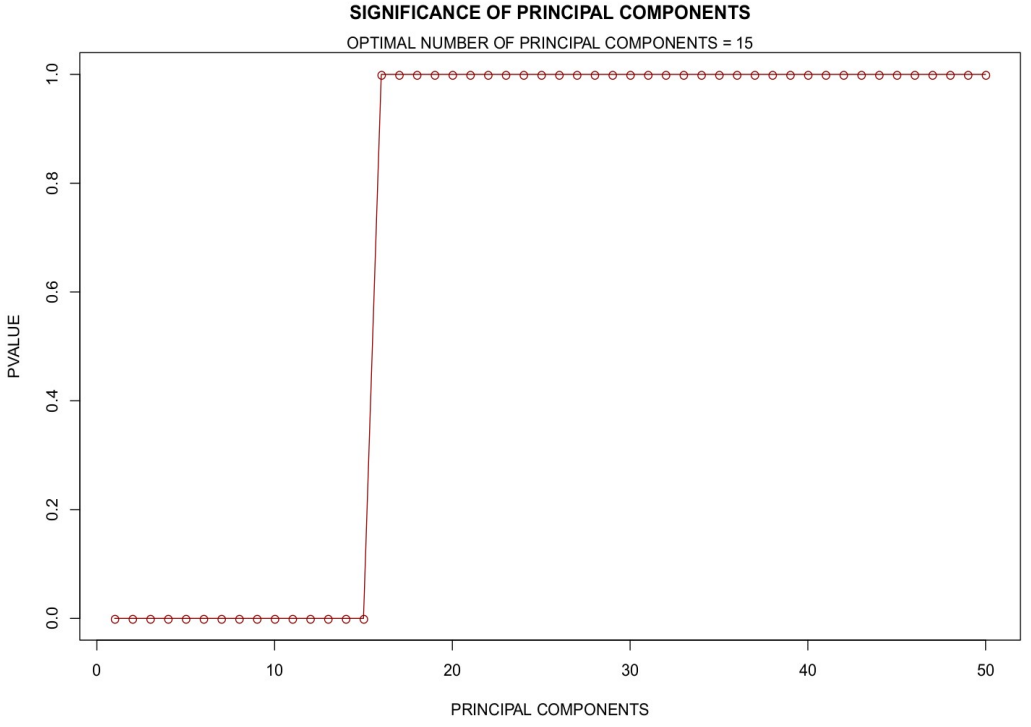


Low Variation in 1000G_EUR Data



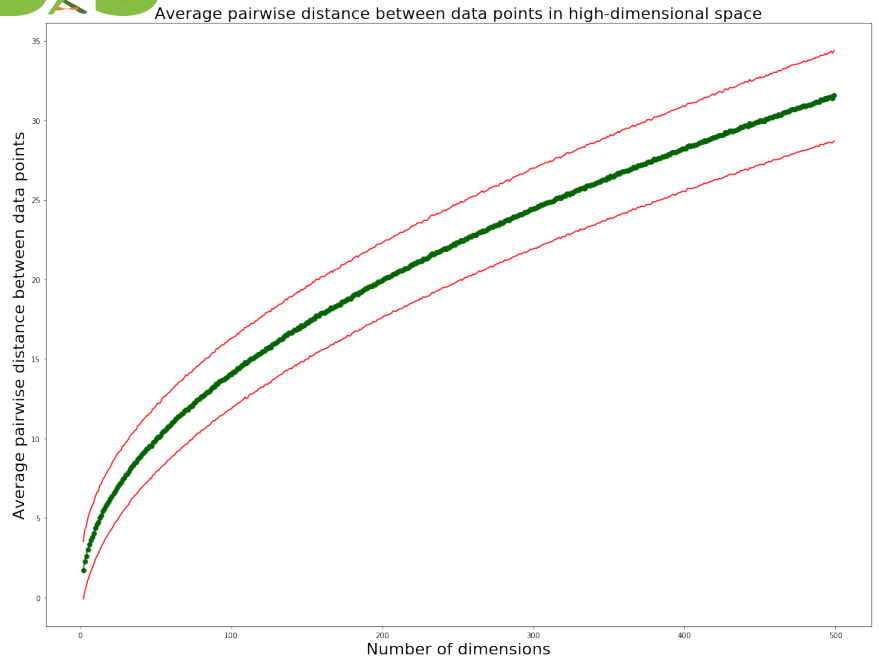
How is Variance Explained in Other OMICs?

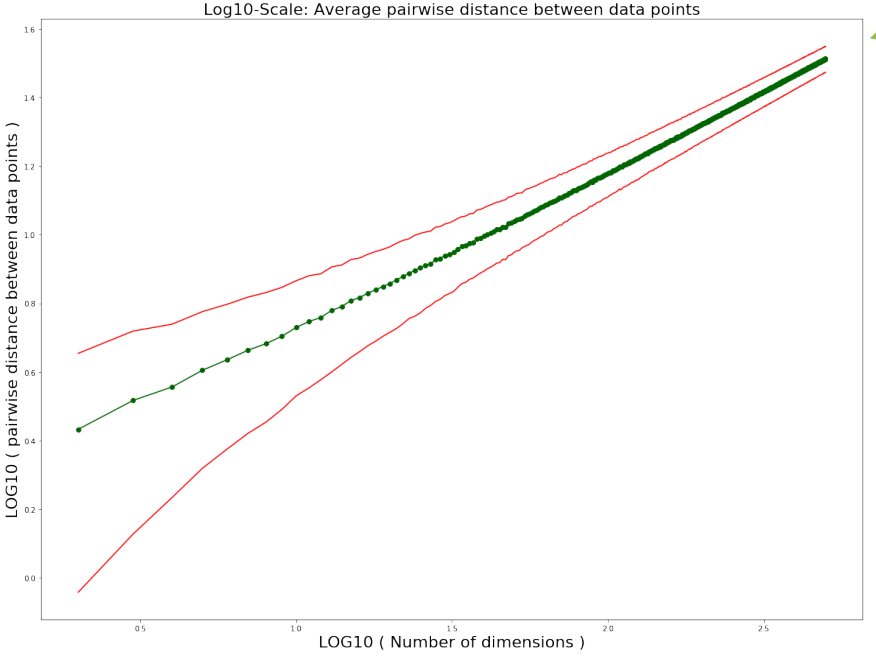




In other OMICs the observed variance explained is typically well separated from the variance explained of the shuffled data matrix

Distances in High Dimensions





Pairwise distances grow as:

dist ~ sqrt (p)

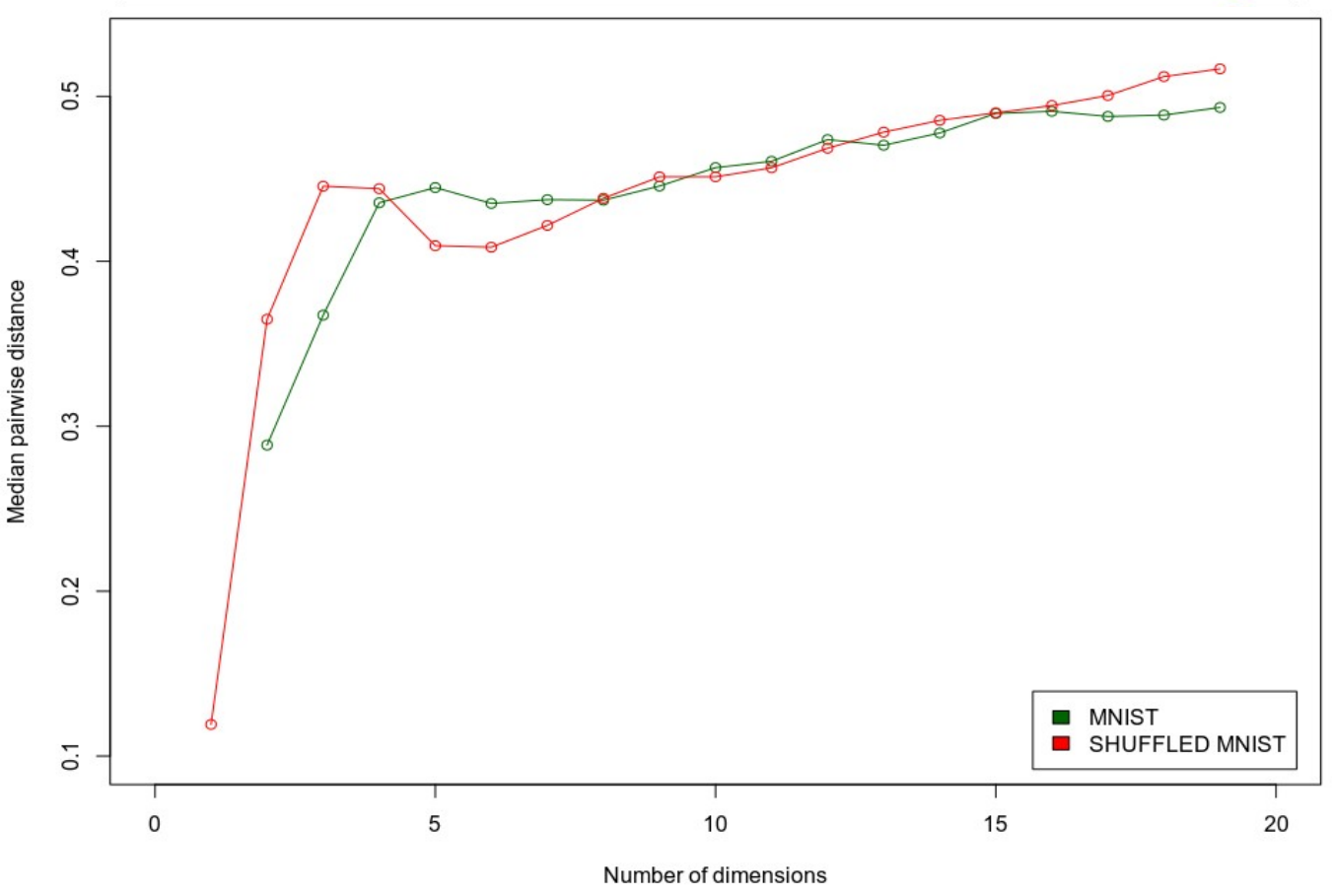
data points become far apart

Dep. Variable:	y	**R-squared (uncentered):**	1.000
Model:	OLS	**Adj. R-squared (uncentered):**	1.000
Method:	Least Squares	**F-statistic:**	1.454e+06
Date:	Mon, 01 Feb 2021	**Prob (F-statistic):**	0.00
Time:	17:48:02	**Log-Likelihood:**	734.17
No. Observations:	498	**AIC:**	-1466.
Df Residuals:	497	**BIC:**	-1462.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
x1	0.5636	0.000	1205.756	0.000	0.563	0.565

Omnibus:	100.625	**Durbin-Watson:**	0.007
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	161.978
Skew:	1.274	**Prob(JB):**	6.71e-36
Kurtosis:	4.145	**Cond. No.**	1.00

Clustering in UMAP Space



MNIST
benchmark dataset



It looks like replacing original data by UMAP components can mitigate the harm from the Curse of Dimensionality and allow better clustering